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Quantum-resilient blockchain framework for privacy-preserving genomic data sharing and analysis

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ABSTRACT

The rapid expansion of Next-Generation Sequencing (NGS) technologies has positioned genomic data at the forefront of personalized healthcare. Yet, its sensitive nature introduces significant challenges related to security, privacy, and integrity—especially in the face of emerging quantum computing threats. This paper presents a quantum-resilient blockchain framework that integrates post-quantum cryptographic primitives, privacy-preserving machine learning, and decentralized data governance to ensure secure sharing and inference of genomic data. Leveraging lattice-based encryption schemes such as CRYSTALS-Kyber and Dilithium within a Hyperledger Fabric-based permissioned blockchain, the system guarantees resistance to quantum adversaries while maintaining confidentiality and auditability of genomic transactions. The architecture employs transformer-based genomic encoders for high-fidelity mutation detection and expression profiling, alongside hybrid data structures (Cuckoo filters and B+ trees) for fast, privacy-preserving variant querying. Smart contracts enforce an incentive-driven reward-penalty model, ensuring fair access while disincentivizing dishonest behavior. Empirical evaluations demonstrate a transaction throughput of 125 transactions per second, an average block formation time of 250 ms, and a post-quantum signature verification latency of 3.8 ms. The system achieved a genomic mutation detection accuracy of 97.31% and reduced query latency by up to 55% compared to existing methods. Additionally, it exhibited 1.75× faster access response times and 40% lower on-chain resource consumption by utilizing optimized off-chain storage and computation. Beyond theoretical design, our framework highlights practical integration pathways: (i) large post-quantum key sizes are managed through hybrid key encapsulation and off-chain storage, (ii) blockchain scalability issues are mitigated via optimized consensus and cross-network channeling, achieving high throughput and low latency, and (iii) quantum protocols such as QKD and Grover-accelerated search are incorporated in a modular fashion to address state preparation and storage challenges. This end-to-end quantum-secure framework offers a scalable, high-performance solution for collaborative genomic research, enabling compliance with emerging data privacy regulations while future-proofing against quantum-era threats.

1. Introduction

Recent advances in high-throughput sequencing and computational genomics have catalyzed a paradigm shift toward precision medicine, enabling early disease detection, risk profiling, and personalized treatment strategies. As genomic databases continue to expand in both scale and importance, secure, efficient, and privacy-preserving data management becomes a critical challenge. Traditional genomic data infrastructures, typically centralized, face inherent limitations, including a single point of failure, unauthorized access risks, and inefficiencies in large-scale querying. The sensitive nature of genomic data demands a robust solution that guarantees data privacy, integrity, and

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transparency, particularly as such information can reveal predispositions to genetic disorders and influence clinical decision-making [1]. In centralized models, access control often proves inadequate and opaque, discouraging collaboration and exposing individuals to privacy risks [2]. Furthermore, traditional relational and hierarchical database systems are poorly optimized for large-scale, unstructured genomic data, resulting in slow query processing and analytic bottlenecks [3].

To address these challenges, we propose a novel framework that integrates quantum-resilient blockchain technologies with secure genomic analytics. Our approach is grounded in three primary innovations: (i) a permissioned, decentralized blockchain infrastructure using post-quantum cryptographic primitives to resist quantum adversaries, (ii) a fine-tuned, context-aware genomic encoder inspired by transformer models to support scalable, privacy-preserving inference, and (iii) a secure access-control mechanism backed by smart contracts and incentive-based fairness protocols.

Unlike conventional cryptographic systems that rely on RSA or ECC, our system adopts lattice-based post-quantum encryption schemes such as CRYSTALS-Kyber and Dilithium to secure genomic datasets. A threshold key-sharing mechanism is employed to distribute trust across computing nodes and repository holders, mitigating the risk of unilateral key compromise [4]. In parallel, genomic data is encrypted, indexed, and stored using a hybrid of Cuckoo filter hashing and B+ tree partitioning to allow secure, high-speed querying.

To complement blockchain's secure storage and auditability, we leverage transformer-based modeling techniques for DNA analysis, treating sequences as contextual biological languages. Our system adapts a pre-trained genomic BERT model with custom encoding strategies for Next-Generation Sequencing (NGS) data, enabling accurate mutation identification and expression profiling [5,6]. These AI-driven inferences are transparently logged to the blockchain ledger, establishing a verifiable history of analytic outcomes for traceability and trust.

By combining privacy-preserving computation, post-quantum cryptographic protocols, and blockchain consensus mechanisms, our framework ensures that data providers retain control over sensitive information while enabling secure research collaboration and AI-enhanced genomic discovery [7]. The system is experimentally validated using real genomic data and evaluated against metrics such as block formation time, signature latency, policy propagation, and attack resilience under simulated quantum adversary conditions. The major key contributions are as follows.

- A novel blockchain-based architecture using lattice-based cryptography (Kyber/Dilithium) for secure genomic data storage and access in a post-quantum threat model.
- Design and implementation of smart contracts for fair, audit able, and incentivized data sharing, including economic deterrents for dishonest behavior using a penalty–reward mechanism.
- Development of privacy-preserving variant lookup using homomorphic encryption, Cuckoo filters, and secure multi-party computation to reduce query complexity.
- Empirical validation on a private Hyperledger Fabric network, demonstrating high transaction throughput, low execution latency, and strong adversarial resistance in a simulated quantum environment.
- A practical integration roadmap combining blockchain, machine learning, post-quantum cryptography, and quantum protocols. We demonstrate how hybrid key encapsulation reduces PQC overhead, lightweight off-chain storage manages signature sizes, and Hyperledger Fabric optimizations overcome low throughput and latency. Additionally, quantum state preparation and storage challenges are addressed through modular simulation environments, bridging theoretical advances with deployment feasibility.

1.1. Paper organization

The remainder of this paper is organized as follows. Section 2 presents a comprehensive review of related literature, focusing on blockchain-based solutions for genomic data privacy, and the application of deep learning — particularly transformer-based models — in genomic analysis. Section 3 provides the necessary background, covering core concepts of blockchain technology, post-quantum cryptographic primitives such as Kyber and Dilithium, secure multi-party computation, and transformer models for biological sequence interpretation. In Section 4, we detail the design of the proposed framework, including the system architecture, smart contract logic, consensus mechanism, and cryptographic data flow. Section 5 describes the threat model, emphasizing resistance to post-quantum adversaries, and presents a formal security analysis based on simulation privacy and economic deterrence formulations. Section 6 outlines the experimental setup, including tools used (Hyperledger Fabric, Open Quantum Safe APIs), datasets, performance metrics, and adversarial simulations. This section also presents the evaluation results with relevant figures and tables. Section 7 discusses the key findings, analyzing performance, scalability, and security under the proposed quantum-resilient architecture. Finally, Section 8 concludes the paper and suggests future research directions, including real-time AI-assisted genomic workflows, federated genomic blockchain systems, and integration with quantum key distribution technologies.

2. Related work

In recent years, the field of genomics has witnessed significant interest in applying blockchain technology to address challenges in data security, privacy, and decentralized governance. Traditional centralized genomic storage models have proven vulnerable to unauthorized access, single points of failure, and compliance issues with cross-border data governance. To address these concerns, various blockchain-based frameworks have emerged, offering immutable, auditable, and decentralized infrastructures for genomic data sharing [5]. Recent studies such as [8,9] demonstrate how decentralized health data systems can incorporate federated analytics and blockchain interoperability, but they still rely on classical cryptography, leaving them vulnerable to quantum-era adversaries.

Several proposals have employed permissioned blockchains with smart contracts to enforce access control policies. For instance, smart contract-enabled systems allow users to define consent conditions, delegate access rights, and manage data visibility across institutions without centralized intermediaries. Projects like Zenome exemplify this approach by combining user-driven data control with monetization options, using hybrid blockchain architectures with both public and private layers [10].

Beyond access control, some architectures propose novel consensus mechanisms tailored to genomic use cases. For example, Proof-of-Genomic-Stake (PoGS) assigns voting rights based on data contribution quality and volume [11]. These domain-specific consensus methods improve fairness but often require tailored trust models and high-performance validation logic. Additionally, scalable block storage has been supported through Merkle Patricia Tries (MPT) and hash-linked data structures, which allow for efficient mutation-level indexing and auditability.

Privacy-preserving data storage and querying techniques have also gained traction. Homomorphic encryption and secure multi-party computation (SMC) enable federated analytics without disclosing raw genomic content. However, these methods face performance bottlenecks, especially when applied at population scale due to their computational intensity [12,13]. To mitigate this, some solutions propose overlapping data segmentation or compressed bloom filters to reduce query complexity, but they trade off between sensitivity and efficiency.

Table 1
A comparative analysis of existing work.

Paper	Objective	Technology used	Drawbacks	Key study
[16]	Propose secure sharing of genomic data using blockchain	Blockchain, Smart Contracts	High computational costs associated with blockchain implementations.	Enhanced privacy and authenticity in data sharing.
[5]	Address interoperability and regulatory issues in genomic data sharing	Consortium Blockchain, Merkle Patricia Trie (MPT)	Complexity in ensuring data compatibility and interoperability.	Solves data format compatibility issues.
[12]	Improve genomic prediction tasks using AI models	BERT (Bidirectional Encoder Representations from Transformers)	Potential bias in models trained on genomic data.	Better results in diverse tasks of genomic prediction.
[14]	Integrate different genomics data for improved prediction	Multimodal Transformers	Challenges in effectively integrating and processing multimodal data.	Optimal performance in gene expression and regulatory element activity prediction.
[14]	Enhance detection of distal regulatory elements	Self-Attention Mechanisms, Position-Aware Attention Modules	Complex mathematical relations for spatial dependencies.	Effective for long-distance genomic sequence analysis.
[17]	Enable joint training of models without sharing key genomics data	Federated Learning, Blockchain	High costs involved in model training and consensus algorithms.	Versatile solutions for data privacy.
[18]	Enhance privacy-preserving mechanisms for genomic data	Blockchain, Homomorphic Encryption	High costs in performing homomorphic operations.	New partitioning strategies for data privacy.
[19]	Address ethical concerns in genomic data handling	Blockchain, Smart Contracts	Challenges in designing changeable methods for consent and data rights.	Recommendations for ethical use of blockchain-based systems.
[20]	Improve epigenomic prediction tasks using T5-based models	T5-Based Transformers, Masked Epigenome Modeling (MEM)	Limited exploration of pre-training objectives.	Perform well in diverse epigenomic prediction tasks.

More recently, quantum-safe blockchain infrastructures have been proposed as a countermeasure to the emerging threat of quantum computing. Classical cryptographic protocols such as RSA and ECDSA are vulnerable to quantum attacks via Shor's algorithm [7]. In contrast, lattice-based schemes like CRYSTALS-Kyber (for key exchange) and Dilithium (for digital signatures) offer resistance against quantum adversaries. These primitives have been incorporated into blockchain layers and smart contracts in various prototypes to future-proof genomic data systems [14]. Latest evaluations [15] further confirm the readiness of lattice-based schemes such as Kyber and Dilithium for integration into blockchain networks. Our framework builds directly on these standardized primitives, but extends their utility by combining them with genomic encoders and incentive-driven smart contracts.

Several quantum-resilient frameworks introduce threshold key sharing and distributed decryption among multiple nodes, ensuring that no single entity can compromise the data or access logs [21]. When combined with blockchain consensus algorithms such as quantum-hardened BFT or post-quantum Proof-of-Stake (PQPoS), these systems preserve decentralization while strengthening integrity under adversarial models [17].

Comparatively, most existing blockchain solutions for genomics still rely on traditional encryption methods and have not yet adopted quantum-safe cryptographic primitives in Table 1. They also lack cross-domain scalability, where genomic datasets must interoperate across jurisdictions, institutions, and research consortiums [22]. Furthermore, consent frameworks in tab many blockchain-based systems are static, with limited support for dynamic consent, revocation, and user autonomy. Addressing these gaps, our framework integrates lattice-based cryptography, cross-domain permissioned channels, and smart contracts for fine-grained consent and query logging. Ethical and legal implications are also an active area of research. Scholars have emphasized the importance of designing systems that support citizens' rights to data erasure and dynamic informed consent [18]. Blockchain-based systems provide an immutable audit trail, but this immutability can conflict with evolving privacy laws like GDPR. Some proposed solutions include reversible commitments and time-locked smart contracts to mediate such conflicts [23].

Therefore, although multiple systems have explored blockchain for genomic data security and privacy, most lack quantum resilience, fine-grained consent, or scalable privacy-preserving query mechanisms. Our proposed framework advances the state of the art by integrating post-quantum cryptographic security, dynamic smart contracts, and scalable off-chain querying for high-throughput genomic applications. n contrast to prior works that either focus narrowly on cryptographic primitives or genomic AI models, our framework integrates blockchain, post-quantum cryptography, transformer-based genomic encoders, and quantum protocols into a unified, practically deployable system.

3. Problem formulation

The exponential growth in genomic data, driven by Next-Generation Sequencing (NGS) technologies, has created unprecedented opportunities for precision medicine, disease risk prediction, and population-scale genetic studies. However, these data are inherently sensitive and personal and pose serious challenges in privacy, integrity, and accessibility. Traditional data protection mechanisms, including centralized access control, public key infrastructure (PKI), and symmetric encryption, are increasingly inadequate in the face of quantum computing threats, such as Shor's algorithm that compromises the RSA and ECC schemes [24].

Let $\mathcal{G} = \{G_1, G_2, \dots, G_n\}$ represent genomic datasets contributed by data providers $\mathcal{P} = \{DP_1, DP_2, \dots, DP_n\}$, where each $G_i \in \mathbb{Z}_4^m$ denotes a genomic sequence encoded over the nucleotide alphabet {A, T, G, C}. These sequences are sensitive and demand secure mechanisms for privacy, integrity, and controlled access—especially under the emerging threat of quantum computing.

The problem lies in enabling encrypted querying and decentralized sharing of specific genomic subsequences $G_i^{[a:b]} \subset G_i$, with positions a, b, without revealing:

- The full genome G_i,
- The query region $[a : b]$,
- The identity or behavior of data clients $DC_j \in \mathcal{C}$.

To address this, we introduce a quantum-secure, blockchain-based framework using the following techniques:

3.1. Quantum key distribution (QKD)

For any two nodes $u, v \in \mathcal{N}$, a shared session key k_{uv}^Q is established using the BB84 protocol, ensuring information-theoretic security against eavesdropping:

$$k_{uv}^Q \xleftarrow{\text{BB84}} \text{QKD}(u, v) \tag{1}$$

3.2. Post-quantum encryption

Each genomic record is encrypted using lattice-based encryption such as CRYSTALS-Kyber, secure against quantum adversaries:

$$\text{Enc}(G_i) = c_i = f_{\text{Kyber}}(G_i, pk_i) \tag{2}$$

where pk_i is a public key based on Ring-LWE hardness assumptions. As shown in Eq. (2), the ciphertext $C = (u, v)$ represents the Kyber-based encapsulated form. In this framework, CRYSTALS-Kyber is adopted as the key encapsulation mechanism due to its NIST PQC standardization and security guarantees based on the hardness of Module-LWE. Unlike FrodoKEM, which produces ciphertexts exceeding 15 KB and introduces impractical storage overhead for blockchain-based systems, Kyber ciphertexts remain in the kilobyte range (≈1.2 KB) with encapsulation/decapsulation times of 1 ms on commodity CPUs. This ensures genomic query encryption remains efficient without introducing prohibitive latency.

3.3. Blockchain-based access logging

Access transactions are logged as immutable entries on a permissioned blockchain $\mathcal{B}$, represented as tuples:

$$A = \{(DC_j, DP_i, t, r)\} \tag{3}$$

ensuring decentralized and tamper-evident audit trails.

3.4. Quantum zero-knowledge proofs (QZKPs)

To validate access rights without revealing the client's identity or query target, a zero-knowledge proof verifies token possession:

$$\text{VerifyZK}(\tau_j, \mathcal{L}) = 1 \quad \Rightarrow \quad \text{Access Granted} \tag{4}$$

where $\tau_j \in \mathcal{T}$ is a quantum-valid token and $\mathcal{L}$ is the on-chain access control ledger.

3.5. Quantum-accelerated variant search

To search for a variant $v \in G_i$, we simulate Grover's algorithm to find the match index $\hat{i}$ with quadratic speedup:

$$G_i[\hat{i}] = v \quad \text{in } \mathcal{O}(\sqrt{n}) \text{ time} \tag{5}$$

This approach provides quantum-resilient, privacy-preserving genomic data access and indexing, ensuring confidentiality, integrity, and regulatory compliance even under quantum adversarial models.

Table 2
Notations and definitions used in this framework.

Notation	Description
Q_t	Cipher text
P_r	Plain text
P_{key}	Public key
r	Coefficient of plain text module
t	Degree of polynomial
S_{key}	Secret key
$S_{\text{key1}}, S_{\text{key2}}$	First and second secret key
a	Γ Integer
P_r, Q_t	Ring of polynomial of both plain and cipher text
Enc(·)	Encoding function
Ψ	Considered error
S_i	Reference size
$\zeta\tau_i$	Partitioning block
B_{TOG}	B+ tree original gene
STC	Straight to client
DP	Data Provider
RN	Repository Node
GTP	Genomic Transaction Entity
PUK	Public Key
PVK	Private Key
DC	Data Client

4. Preliminaries and emerging technology

The previous knowledge of blockchain, full encryption, bloom filters, and genomics required to comprehend our suggested structure is covered in this part of the paper. The definitions and important notations used in this article are listed in Table 2.

4.1. Genomics data

An entire bundle of human deoxyribonucleic acid (DNA), which is divided into two identical DNA threads, is known as the human genome [25,26]. It comprises around twenty-three sets of chromosomes and three billion base pairs. Adenine (A), thymine (T), guanine (G), and cytosine (C) are the bases of DNA; A is associated with G, and T is paired with C. Variations in genes account for the remaining 0.1% of nucleotides, which are identical in 99.9% of cases between any two individuals. Single-nucleotide polymorphism is the most prevalent genetic variation, occurring at a single place in a DNA sequence [1]. Genetic sequence variants can be stored in various file formats and requirements. The variation call format, which consists of a header section and body, is commonly used.

4.2. Blockchain

Blockchain operates as a decentralized ledger, ensuring the integrity and security of transactions within a distributed peer-to-peer (P2P) network [27]. Transactions are grouped into blocks, which undergo validation by network nodes using a consensus mechanism [27] Validated blocks are then linked cryptographically to the preceding block in the chain via hash pointers. Each block comprises a block header containing essential metadata, including cryptographic hashes representing the block's data and the current timestamp indicating the time of creation, along with other relevant information. The block body stores transaction data. There are many kinds of blockchain networks, including private-ledger, public-ledger, and consortium ledger [1,24]. These networks find application across sectors such as finance and healthcare, facilitating trustless transactions among participating entities [28]. Rules within these networks are enforced through self-executing codes stored immutably on the blockchain. These codes automatically execute based on predefined guidelines outlined in a collaborative framework. In this initiative, we establish a private blockchain network to regulate interactions between Data Providers (DPs) and Data Clients (DCs).

4.2.1. Key creation

In this section, we define the lattice-based cryptographic primitives used to protect genomic data against quantum attacks. We adopt a key encapsulation mechanism (KEM) inspired by Ring-LWE-based post-quantum cryptography, such as CRYSTALS-Kyber [29].

Let $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n + 1)$ denote the ring of polynomials with coefficients in $\mathbb{Z}_q$, reducing modulo the polynomial $x^n + 1$, where q is a large prime modulus.

To generate the pair of quantum resistant keys, we begin by sampling a secret key $s \in \mathcal{R}_q$ from a discrete binomial or centered Gaussian distribution. A public seed ρ is selected to deterministically generate a matrix $A \in \mathcal{R}_q^{k\times k}$, which serves as part of the public key. An error vector $e \in \mathcal{R}_q$, drawn from the same distribution, is added to introduce noise.

The public key is then computed as

$$t = A \cdot s + e \mod q, \tag{6}$$

where $t \in \mathcal{R}_q^k$. The resulting key pair is defined as the public key $pk = (A, t)$, and the secret key $sk = s$.

To make the lattice-based key material introduced above practical for real deployments, the system applies a hybrid KEM-DEM strategy. CRYSTALS-Kyber is employed exclusively as the key encapsulation mechanism to establish short-lived symmetric AES-256 session keys, which then protect bulk data transmission. Only the small encapsulation record and a SHA3-256 digest of the ciphertext are stored on-chain, while the actual ciphertext is maintained in the secure off-chain repository. For signature operations, CRYSTALS-Dilithium is used for entity attestation, but verification is batched by endorsers and only the aggregated verification digest is written to the ledger. This selective storage approach avoids burdening the blockchain with multi-kilobyte objects, ensures interoperability with existing Fabric peers, and localizes heavy cryptographic artifacts to secure side channels and repositories while maintaining full post-quantum security guarantees.

4.2.2. Encryption (Encoding)

Given a message $m \in \{0,1\}^n$, representing for example a genomic sequence block, and the public key $pk = (A, t)$, the encryption process proceeds as follows.

First, ephemeral randomness is introduced by sampling $r, e_1, e_2 \in \mathcal{R}_q$, where these elements are drawn from a centered binomial or discrete Gaussian distribution [30]. The binary message m is then encoded into a ring element $\mu \in \mathcal{R}_q$, using an appropriate embedding function.

With these components, the ciphertext is computed using the following equations:

$$\begin{aligned} c_1 &= A^T \cdot r + e_1 \mod q, \\ c_2 &= t^T \cdot r + e_2 + \mu \cdot \left\lfloor \frac{q}{2} \right\rfloor \mod q. \end{aligned} \tag{7}$$

The final ciphertext is represented as the tuple $ct = (c_1, c_2)$, where both components lie in $\mathcal{R}_q$ and are suitable for secure transmission or storage.

4.2.3. Decryption (Decoding)

To decrypt a ciphertext $ct = (c_1, c_2)$ using the corresponding secret key $s \in \mathcal{R}_q$, the receiver first computes an intermediate ring element $\hat{\mu} \in \mathcal{R}_q$ as follows:

$$\hat{\mu} = c_2 - s^T \cdot c_1 \mod q. \tag{8}$$

This operation effectively removes encryption randomness and recovers the encoded message component [31]. To retrieve the original binary message $m \in \{0,1\}^n$, the result is scaled and rounded to the nearest bit using a decoding function.

$$m = \text{Decode}\left(\left\lfloor \frac{2 \cdot \hat{\mu}}{q} \right\rceil \right) \tag{9}$$

where $\lfloor \cdot \rceil$ denotes the rounding in components to the nearest integer.

The correctness of the decryption depends on the noise terms e_1, e_2 being sufficiently small so that $\hat{\mu}$ remains close to its expected value modulo q. If the error bounds are satisfied, the message m is successfully recovered. For digital signatures, CRYSTALS-Dilithium is employed. Dilithium offers strong existential unforgeability under chosen message attacks (SUF-CMA) while producing signatures of ≈2.7 KB, which remain compatible with Hyperledger Fabric's block size constraints. By contrast, Falcon, though compact, requires floating-point Gaussian sampling that complicates deployment on standard genomic computing clusters.

4.2.4. Partially decryption

In collaborative or federated genomic data processing environments, it is often necessary to prevent any single party from fully decrypting sensitive data [32]. To address this, we incorporate a *threshold partial decryption* scheme using lattice-based, post-quantum secure primitives.

Let the ciphertext $ct = (c_1, c_2)$ be encrypted under a secret key $s \in \mathcal{R}_q$. We divide the secret key into k additive shares:

$$s = s_1 + s_2 + \cdots + s_k \mod q \tag{10}$$

where each $s_i \in \mathcal{R}_q$ is securely distributed to a distinct authorized party P_i.

Each party independently computes a partial decryption share.

$$\mu_i = c_2 - s_i^T \cdot c_1 \mod q. \tag{11}$$

Once all k shares are computed, they are securely aggregated:

$$\hat{\mu} = \sum_{i=1}^{k} \mu_i \mod q. \tag{12}$$

The original message $m \in \{0,1\}^n$ is then recovered by applying the decoding function:

$$m = \text{Decode}\left(\left\lfloor \frac{2 \cdot \hat{\mu}}{q} \right\rceil \right). \tag{13}$$

This threshold-based partial decryption is secure under the Ring-LWE assumption and prevents any subset of fewer than k parties from reconstructing the message [33]. Moreover, it provides resistance against quantum adversaries by using quantum-secure key material and ensures privacy in collaborative genomic data workflows without central trust.

4.2.5. Cuckoo filter

A Cuckoo filter consists of an array of buckets, each holding a small fingerprint of inserted elements. It uses two hash functions to map each element to two possible locations. On insertion, if both locations are full, the filter relocates elements ("kicks out") recursively until a spot is found or a threshold is reached [34] .

To enable privacy-preserving variant lookup without disclosing full genomic content, we incorporate a Cuckoo filter as a compact data structure to store hashed variant identifiers. For each genomic data set G_i, a Cuckoo filter $\mathcal{F}_i$ is constructed to support fast queries of the form:

$$\text{Query}(v) \Rightarrow \begin{cases} \texttt{True} & \text{if } v \in G_i \\ \texttt{False} & \text{(with probability } 1-\epsilon\text{) if } v \notin G_i \end{cases} \tag{14}$$

where ϵ is the configured false positive rate. The filter can be queried in encrypted form using homomorphic operations or MPC to maintain privacy under quantum adversaries.

5. Proposed framework

In this section, we discuss the proposed system framework depicted in Fig. 1, along with the system model and all the assumptions that we consider to configure our proposed system.

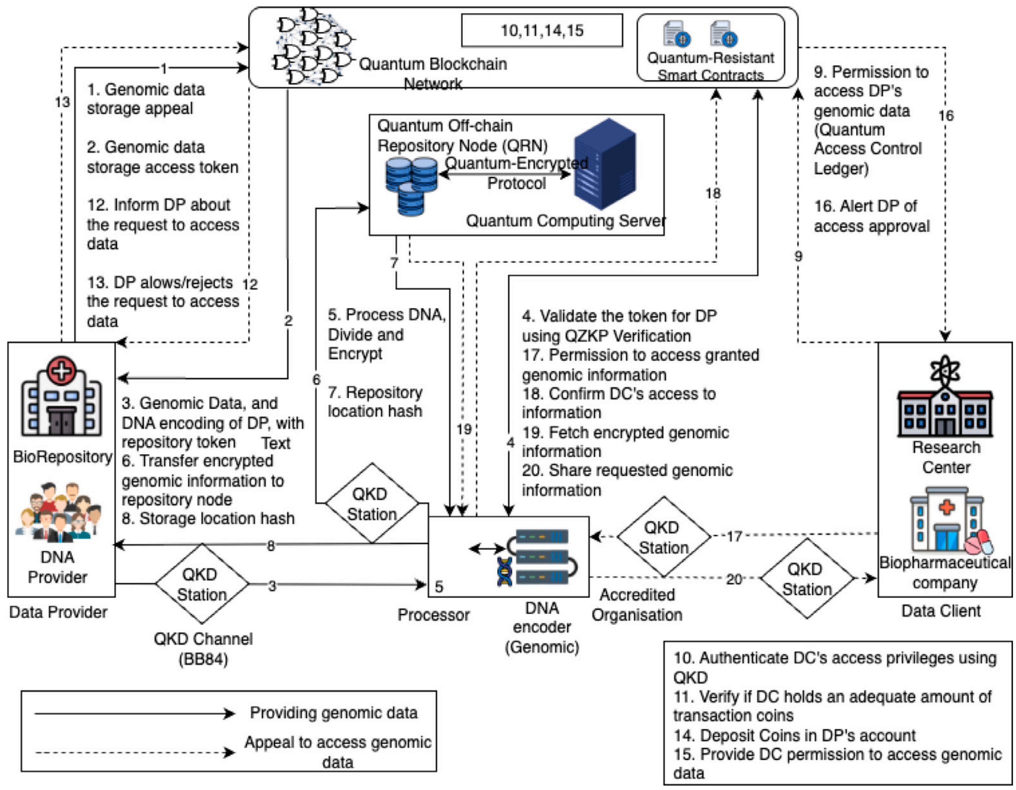


Fig. 1. The blockchain architecture for securely sharing and receiving genomic data.

5.1. System architecture

•Genetic Data Custodian (GDC): The GDC is responsible for securely storing and managing genetic data obtained from providers. They ensure compliance with data protection regulations and facilitate controlled access to authorized parties [35].

•Genetic Data Analyst (GDA): The GDA specializes in analyzing genetic data to extract meaningful insights and correlations. They employ advanced computational techniques to identify patterns and associations within genomic datasets.

•Genetic Data Exchange Platform (GDEP): The GDEP serves as a centralized hub for sharing and exchanging genetic data among research institutions, healthcare providers, and other stakeholders. It provides standardized protocols for data interoperability and facilitates collaborations in genomic research [36,37]. To facilitate real-world deployment, the repository nodes are designed to be compatible with existing healthcare data standards, such as HL7 FHIR (Fast Healthcare Interoperability Resources) for structured health data exchange and DICOM-genomics extensions for integrating genomic and imaging records. This ensures that genomic transactions on the blockchain can interface with widely adopted EHR platforms without requiring disruptive changes to existing hospital IT systems.

•Genomic Data Provider (DP): The DP represents individuals or organizations contributing genomic data for various purposes. They may submit DNA samples or pre-sequenced genomic data to authorized institutions for processing and dissemination [38].

•Genomic Data Client (DC): The DC encompasses entities such as research institutions or medical facilities seeking access to genomic data for analysis and research endeavors [39]. DCs compensate DPs for data access using digital currency or other mutually agreed-upon means.

•Off-chain Repository Node (RN): Due to the challenges associated with storing data on the blockchain, an RN is employed for efficient off-chain data storage. This node collaborates with the blockchain network to ensure secure and accessible storage of genomic data.

•Blockchain Framework (BF): The BF serves as the decentralized platform facilitating transactions and data management within the genomic data ecosystem [40]. It provides access control features for data contributors and implements penalty mechanisms by recording immutable transactions such as data uploads and access requests.

•Computing Server (CS): The CS offers computational services necessary for analyzing genomic data while upholding data privacy and security. It collaborates with storage providers to perform computations on encrypted genomic data.

5.2. Dataset

The NCBI Gene Expression Omnibus (GEO) [15] is a widely used public repository that serves as a comprehensive resource for genomic and functional genomic studies. It hosts a vast collection of high-throughput gene expression datasets, including microarray and RNA sequencing (RNA sequencing) data generated from various organisms, experimental conditions, and disease states. GEO follows MIAME-compliant standards, ensuring that datasets are well-annotated and suitable for reproducible research. In genomic data analysis, GEO is particularly valuable for tasks such as differential gene expression analysis, transcriptome profiling, and the development of machine learning models for biomarker discovery or disease classification. Its accessibility and rich metadata make it an essential platform for researchers looking to explore large-scale expression patterns and uncover underlying biological mechanisms.

5.3. System modeling

The proposed system establishes a quantum-resilient, secure architecture for decentralized genomic data sharing and querying. Integrates quantum key distribution (QKD), postquantum encryption, blockchain-based access control, and quantum zero-knowledge proofs (QZKPs) into a unified framework. The core entities include the Cryptographic

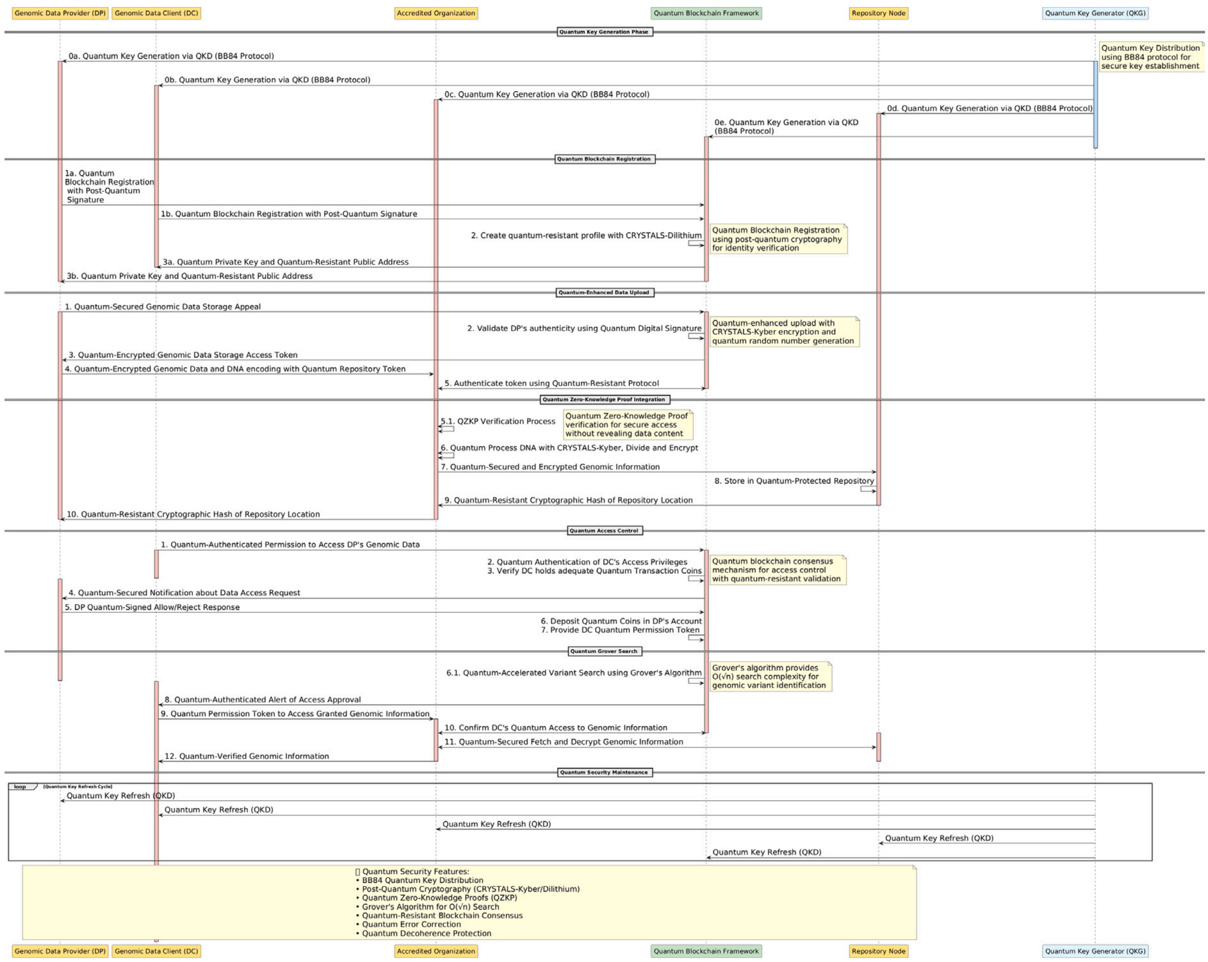


Fig. 2. The sequence of operations in our suggested system for the secure exchange of genomic data via blockchain technology.

Generator (CG), Accredited Organization (AC), Data Providers (DPs), Repository Nodes (RNs), Computing Servers (CSs) and Data Clients (DCs), all interacting through a private Blockchain Framework (BF) as illustrated in Fig. 2.

5.3.1. Cryptographic key generation and distribution

The system begins with the Cryptographic Generator (CG), which is responsible for generating cryptographic keys essential for secure data processing and transmission. Specifically, CG generates a public–private key pair (PUK, PVK), where the public key is used for encryption and the private key for decryption [41]. To improve security, the private key PVK is further divided into two randomized shares PVK_1 and PVK_2 using a verifiable secret sharing scheme, as also adopted in. These components are securely disseminated: the public key PUK is delivered to the Accredited Organization (AC), while PVK_1 is provided to the Repository Node (RN), and PVK_2 is sent to the Computing Server (CS). This distribution shown in Fig. 3 ensures that no single entity possesses the complete decryption authority, thus mitigating risks of internal breaches.

5.3.2. Enrollment of genomic transaction entities

To participate in the blockchain-enabled genomic data exchange, both Genomic Transaction Entities (GTEs) and Data Clients (DCs) undergo a secure registration process into the Blockchain Framework (BF). Upon successful enrollment, each entity is issued a unique pseudonymous blockchain address and a corresponding quantum-secure private key, e.g., derived from CRYSTALS-Dilithium or similar NIST-standardized post-quantum schemes [42]. These keys allow users to perform digital signatures and interact securely with smart contracts. The registration transaction is recorded immutably on the BF, establishing accountability and enabling future verifiable audits while preserving anonymity.

5.3.3. Genomic data upload to the blockchain

Following registration, the Accredited Organization (AC) receives genomic samples, such as those from blood or hair follicles, along with associated phenotypic data from Data Providers (DP). The AC processes this data by first dividing the genomic sequences into multiple fragments or nodes. Each node is individually hashed using a Cuckoo filter to allow efficient set membership checks while preserving privacy [43]. The processed DNA is then encoded and encrypted using lattice-based post-quantum encryption schemes such as CRYSTALS-Kyber, secure against quantum adversaries. After encryption, genomics was performed.d.

5.3.4. Splitting up and processing the original genome

During the system initialization phase, the Control Infrastructure (CI) performs a one-time preprocessing step to divide the original genomic sequence (OG) into non-overlapping segments and indexes the individual node keys using a B+ tree data structure B + for efficient access and retrieval. Given that sequenced genomes in Fig. 4 span multiple chromosomes, locating specific variants on a chromosome is computationally intensive. To alleviate this, we utilize a reference genome during the DNA sequencing process to accelerate variant searches. In our proposed method, reference genomes such as

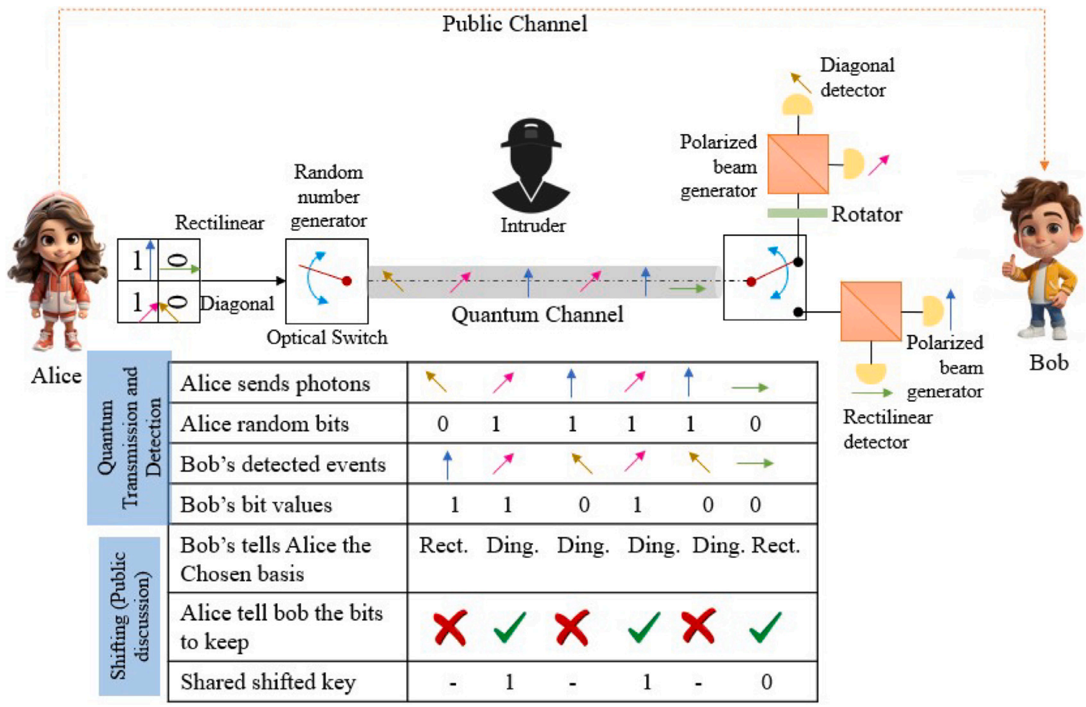


Fig. 3. Key Distribution.

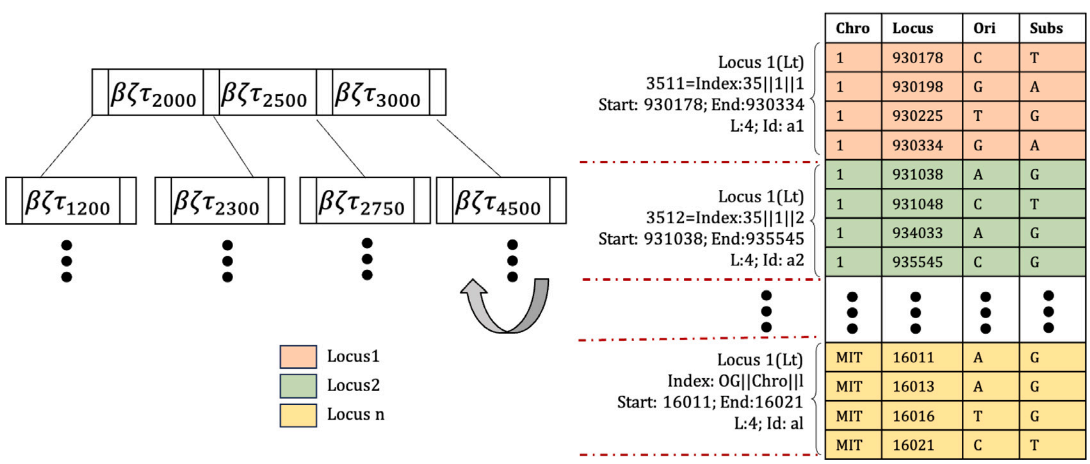


Fig. 4. The B+ tree partitioning of the genomic data for fast retrieval and storing.

`NCBI34`, `NCBI35`, `NCBI36`, `GRCh37`, and `GRCh38` are partitioned into disjoint chromosomal regions. Each partition block is then transformed into a B+ tree structure, significantly reducing the search space required for the detection of secure genomic variants.

Let the size of the original reference genome be denoted by S_i, where $i \in \{1, 2, 3, \dots, \zeta\}$, and ζ represents the total number of partitions. For each segment ζ_τ^i, an index value $\zeta_\tau^{(i.\text{index})}$ is computed as:

$$\zeta_\tau^{(i.\text{index})} = \frac{|\texttt{OG}|}{|\texttt{Ch}|} \cdot i \tag{15}$$

Here, $|\texttt{OG}|$ denotes the length of the original genome, and $|\texttt{Ch}|$ is the number of chromosomes.

Upon computing the segment index, a corresponding code $\zeta_\tau^{(i.\text{code})}$ is generated. These two values — $\zeta_\tau^{(i.\text{index})}$ and $\zeta_\tau^{(i.\text{code})}$ — are used to construct a B+ tree, as illustrated in Fig. 4, to facilitate fast data partitioning and retrieval. We proposed a search function as $|\texttt{OG}|$. It introduces a novel search function called `B_TOG` for efficient querying within the original genome:

$$\texttt{B\_TOG}(g_{\zeta_i}) = \texttt{key} = g_{(\zeta_i.\text{index})}, \quad \texttt{outcome} = (\zeta_\tau^{(i.\text{code})},\ \zeta_\tau^{(i.\text{start})},\ \zeta_\tau^{(i.\text{end})},\ \zeta_\tau^i) \tag{16}$$

where: $\zeta_\tau^{(i.\text{start})}$: Starting position of the partition block $\zeta_\tau^{(i.\text{end})}$: Ending position of the partition block ζ_τ^i: The size of the partition

For example: TP53 Partitioning. To illustrate, consider the partitioning of the `TP53` gene. Assume the node starting position is 1 and the block size is 4. The computed values are:

$$g_{\zeta_i} = \text{key} = \zeta_\tau^{(i.\text{index})} = 3512$$

$$\text{outcome} = (\zeta_\tau^{(i.\text{code})} = 83248131,\ \zeta_\tau^{(i.\text{start})} = 930178,\ \zeta_\tau^{(i.\text{end})} = 930334,\ \zeta_\tau^i = 4)$$

It is important to note that constructing the `BT_RG` (B+ Tree for Reference Genome) and partitioning the reference genome are one-time tasks. They are updated only when the Genome Reference Consortium releases new benchmark updates.

5.4. Decentralized consequence approach

We suggest a new way to penalize people for not being honest using blockchain technology. This method uses digital coins to reward people who are honest and punish those who are not. We focus on catching dishonest actions like pretending to be someone else or uploading fake genetic information [43,44]. Our system makes sure that everyone is encouraged to be honest most of the time. The computation cost of honest people is denoted by $\mathrm{ComH}_{\mathrm{cost}}$ and for dishonest ones as $\mathrm{ComM}_{\mathrm{cost}}$. The possibility of honest behaviors is pr_h, then dishonest chances are $\mathrm{pr}_m = \mathrm{pr}_{\mathrm{check}}(1 - \mathrm{pr}_h)$, where $\mathrm{pr}_{\mathrm{check}}$ is the checking parameter of dishonest data providers. χ is the reward for the accurate approach and ρ is the penalty for misconduct.

PenaltyRewardDEE (PRDEE) is an algorithm crafted to uphold integrity within data entry procedures, achieving this by penalizing deceptive Data Entry Entities (DEEs) and rewarding trustworthy ones. Commencing with the retrieval of the deceptive DEE ($\mathrm{DEE}_{\mathrm{DE}}$) from the DEE Enrollment Registry (DEER) through address mapping, PRDEE ensures the entity's existence and valid identification. If $\mathrm{DEE}_{\mathrm{DE}}$ meets these criteria, its malicious points are incremented by 1, leading to its subsequent removal from the blockchain framework (BF). The algorithm then conducts an iterative process through DEER to discern trustworthy DEEs, forming a set (H) comprised solely of these entities. Each trustworthy DEE undergoes payoff calculation, which involves dividing the total processing fee by the count of trustworthy DEEs in set H. This calculated payoff is then disbursed to each DEE within set H, accompanied by an increment in their respective reward points. PRDEE concludes its operations by returning 1 to signify successful execution.

PRDEE's functionality can be expressed through the mapping operation for DEER, wherein DEE addresses are associated with their corresponding DEE objects: $\{\ \mathrm{add}_{\mathrm{DE1}} : \mathrm{DEE}_1, \mathrm{add}_{\mathrm{DE2}} : \mathrm{DEE}_2, \dots\ \}$. Furthermore, the set of trustworthy DEEs (H) can be represented as a subset of DEER: $\{\ \mathrm{DEE}_1, \mathrm{DEE}_3, \dots\ \}$. Payoff calculation involves dividing the total processing fee ($\$F_{\mathrm{processing}}$) by the count of trustworthy DEEs in set H, expressed as $\$F_{\mathrm{payoff}} = \frac{\$F_{\mathrm{processing}}}{\mathrm{count}(H)}$

Before any service, such as uploading or browsing genomic data can be used, a Data Exchange Entity (DEE) must be registered on the blockchain in our system. In addition, this procedure entails adding DEE's to the registry of active DEEs (DEERs), as well as producing unique public addresses for each entity [45]. This address is very important in verifying transactions before they take place and also it helps in giving rewards or penalties depending on how the DEEs have behaved [46,47]. The following steps are considered when conducting an enrollment process:

- Step 1: When the commencement of enrollment is initiated by the DEE submitting an enrollment request to the blockchain, then followed by creating an account for him/her with a unique ID based on the current registry of active DEEs. To avoid duplication during registration, if a "new" identifier already exists, meaning that this particular individual has been registered earlier, then there would be an error encountered.
- Step 2: For DEEs not previously registered on the blockchain, the count of registered DEEs is incremented by one and a unique ID is assigned to the new DEE. Then, the address of the DEE, its user type, malicious points (initially zero), and reward points (initially zero) are updated in the DEER.
- Step 3: After that, a new DEE becomes an active participant in our system who is documented within the blockchain ledger as an enrollment transaction, whereupon it receives a success notification.

With these secure measures in place, genomic data transfer to the blockchain can occur in a series of controlled steps. In this regard, **Algorithm 3**, called `TransferingGenomicData (TGD)`, oversees this process, ensuring data integrity and accuracy. The steps involved are as follows:

Algorithm 1 Quantum-Secure Genomic Data Provider Penalty-Reward (QSGDP-PR) Algorithm: Penalizes malicious genomic data providers by redistributing their staked tokens to trustworthy providers

Input: $addr_{DP}, S_{\mathrm{stake}}, \mathcal{B}$ (blockchain ledger)
Output: `Bool` (0 - false, 1 - true)
Retrieve malicious data provider from enrolled providers registry $\mathcal{P}$
$DP_m \leftarrow \mathcal{P}[addr_{DP}]$
if $DP_m.id \le 0$ **or** $DP_m \notin \mathcal{P}$ **then**
    throw InvalidDataProviderException
end if
Verify quantum signature authenticity using post-quantum verification
valid $\leftarrow$ VerifyPQSig(DP_m.signature, $DP_m.\mathrm{pk}_{\mathrm{Kyber}}$)
if valid $= 0$ **then**
    throw InvalidQuantumSignatureException
end if
Increase malicious behavior score and revoke genomic data access
DP_m.malicious_score $\leftarrow DP_m$.malicious_score $+ 1$
DP_m.access_revoked $\leftarrow 1$
Revoke quantum keys and update QKD network
RevokeQKD($DP_m, \mathcal{N}$)   ▷ Revoke quantum keys from network
Filter trustworthy providers with zero malicious score
$\mathcal{T} \leftarrow \{DP_i \in \mathcal{P} : DP_i.\mathrm{malicious\_score} = 0 \wedge DP_i.\mathrm{active} = 1\}$
Redistribute stake among trustworthy providers
$S_{\mathrm{reward}} \leftarrow S_{\mathrm{stake}}/|\mathcal{T}|$
for all trustworthy provider $DP_j \in \mathcal{T}$ **do**
    Execute quantum-secure token transfer
    QSecureTransfer(DP_j.address, S_{reward})
    Update trustworthiness score and blockchain record
    DP_j.trust_score $\leftarrow DP_j$.trust_score $+ 1$
    $\mathcal{B} \leftarrow \mathcal{B} \cup \{(DP_j, \mathrm{REWARD}, t_{\mathrm{now}}, S_{\mathrm{reward}})\}$
end for
Log penalty transaction on blockchain with quantum-resistant hash
$\mathcal{B} \leftarrow \mathcal{B} \cup \{(DP_m, \mathrm{PENALTY}, t_{\mathrm{now}}, S_{\mathrm{stake}})\}$
return 1

Step 1: A Data Provider (DP) generates a request with parameters that include their address add_{DP}, list of reference point hashes $[\mathrm{RP}_{\mathrm{hash}}]'$, authentication fee $\$F_{\mathrm{auth}}$, and a storage token $\mathrm{TKN}_{\mathrm{storage}}$. The system retrieves the DP's unique ID from the blockchain-based identification system known as the *DEE Registry* (DEER). This is achieved by linking the DP's address to the unique identifier. If no valid registration is found or an invalid ID is provided, the process is terminated with an error message.

Step 2: The storage token is then verified for authenticity and validity. This step ensures that the token is legitimate and has not expired.

Verify Storage Token:

If $\mathrm{TKN}_{\mathrm{storage}}$ is invalid or presentTime $> TS_e$, then an error is thrown and the transaction is aborted.

Step 3: Upon successful verification of the storage token, the system retrieves the genomic data item associated with the DP's address from the database of registered genomic transactions.

Retrieve Genomic Data:

$$\mathrm{GDitem} \leftarrow Q[\mathrm{add}_{DP}] \tag{17}$$

Step 4: The transferred genomic data attributes are then allocated. If the provider information is null, it is updated with the DP's address. The genomic data item is then populated with the provider's address, the list of reference points, the authentication cost, and any other relevant attributes.

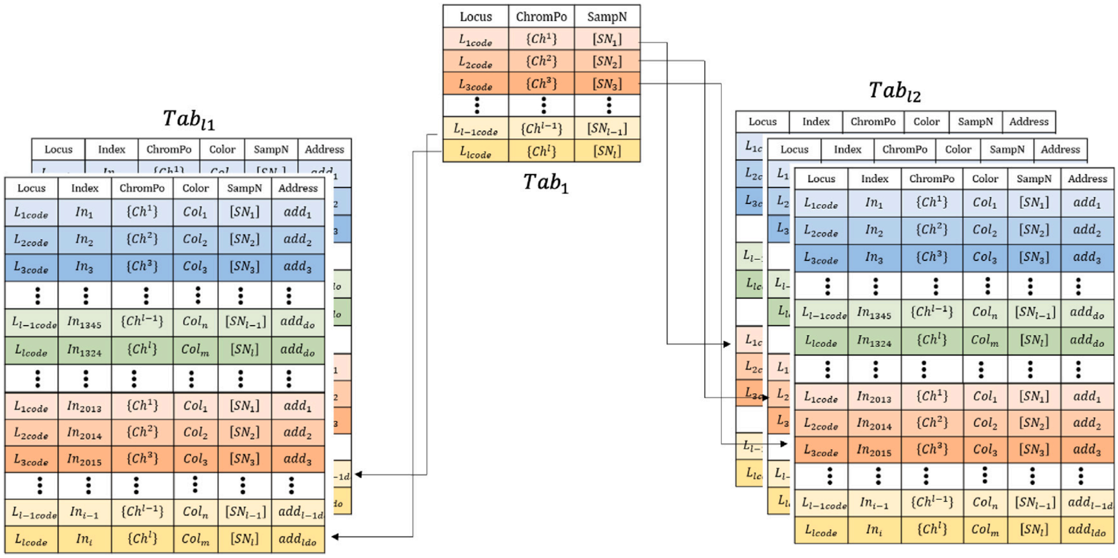


Fig. 5. Genomic category detection and mapping tables.

Update Genomic Data:

$$\text{If GDitem.Provider} = \text{null, then set GDitem.Provider} \leftarrow \text{add}_{DP} \tag{18}$$

Allocate Attributes:

$$\text{GDitem} \leftarrow \{\text{Provider} = \text{add}_{DP},\ RP = [\text{RP}_{\text{hash}}]',\ \text{cost}_{\text{amount}} = \$F_{\text{auth}},\ AR = \text{null}\} \tag{19}$$

Step 5: Finally, the algorithm returns a success indicator (true) if all steps are completed without errors, confirming that the genomic data has been successfully transferred and recorded onto the blockchain framework, considered as success and value=1.

Algorithm 2 Quantum-Secure Genomic Data Provider Enrollment (QSGDP-E)

Input: $addr_{entity}, \mathcal{K}_{\mu}, pk_{Kyber}$
Output: `Bool` (0 - false, 1 - true)
$id_{entity} \leftarrow \mathcal{P}[addr_{entity}].id \cup \mathcal{C}[addr_{entity}].id$
if $id_{entity} > 0$ **then**
  throw AlreadyEnrolledException
end if
if VerifyKyberPK(pk_{Kyber}) $= 0$ **then**
  throw InvalidPostQuantumKeyException
end if
$\mathcal{K}_{QKD} \leftarrow \{\}$
for all $e \in \mathcal{N}$ **do**
  $k^{Q}_{entity,e} \xleftarrow{\text{BB84}} \text{QKD}(addr_{entity}, e.addr)$
  $\mathcal{K}_{QKD} \leftarrow \mathcal{K}_{QKD} \cup \{k^{Q}_{entity,e}\}$
end for
$id_{entity} \leftarrow EntityCount + 1$
if $\mathcal{K}_{\mu} = \text{DATA\_PROVIDER}$ **then**
  $\mathcal{P}[addr_{entity}] \leftarrow DP(id_{entity}, addr_{entity}, pk_{Kyber}, \mathcal{K}_{QKD}, 0, 0, 0)$
else if $\mathcal{K}_{\mu} = \text{DATA\_CLIENT}$ **then**
  $\mathcal{C}[addr_{entity}] \leftarrow DC(id_{entity}, addr_{entity}, pk_{Kyber}, \mathcal{K}_{QKD}, 1, \{\})$
end if
$\pi_{enroll} \leftarrow \text{GenerateQZKP}(addr_{entity}, pk_{Kyber}, \mathcal{L})$
$\mathcal{B} \leftarrow \mathcal{B} \cup \{(addr_{entity}, \text{ENROLL}, t_{now}, \mathcal{K}_{\mu}, \pi_{enroll})\}$
$\mathcal{N} \leftarrow \mathcal{N} \cup \{addr_{entity}\}$
return 1

A request for authorization of genomic data is initiated by a Data Client (DC). Before sending the authorization request, the DC ensures that it is not the provider of genomic data. The process is as follows in Fig. 6.

Step 1: The DC sends an authorization request with the Data Provider's (DP) address and the authorization fee to the blockchain system by invoking the Request Genomic Authorization (RGA) function. The request includes the addresses of both the DP and the DC, the authorization fee, and the DC's access control token.

Step 2: The system performs the following checks:

- Verifies that the DC is not the same as the DP
- Validates the DC's digital signature

If DC = DP, the request is immediately terminated. If the signature is invalid, a monetary penalty is imposed on the DC.

Step 3: The system checks the validity of the access control token $\Gamma^{ac}_{\text{add}_{DC}}$. If the token is invalid, the DC is penalized and the request is canceled.

Step 4: The system retrieves the genomic data item associated with the DP's address from the genomic items mapping shown in Fig. 5 and assigns it to:

$$\text{GDitem} \leftarrow \text{GenomicItems}[\text{add}_{DP}] \tag{20}$$

Step 5: The system retrieves and updates the access list item of `GDitem` for the DC. The process is as follows:

- If the provider of `GDitem` is not null, retrieve the access record AR for the DC
- If AR.SA $\notin$ {Request, Grant }, then:
  - Check whether the DC's balance F_L is sufficient to cover the cost and the deposit
  - If insufficient, terminate the process

Step 6: The system updates the access state AR.SA to `AuthReqSt.Request`. Additionally:

- The processing fee is set to the deposit amount
- The authorization fee is calculated as $F_{\text{auth}} = F_L - \text{deposit}$

Step 7: The system generates a notification to alert the DP of the requested authorization. The notification includes the transaction number, the addresses of the DP and DC, and the authorization fee.

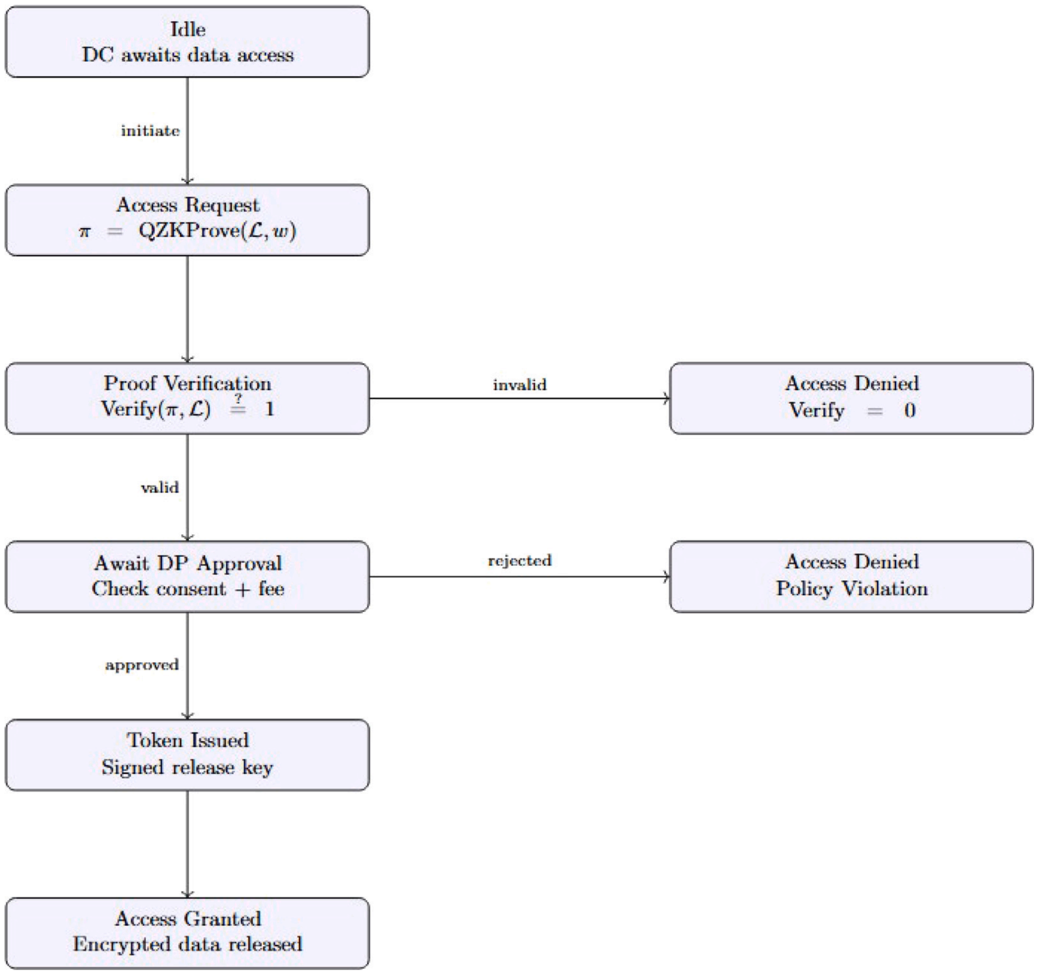


Fig. 6. Workflow.

The **GenomicAccessManager (GAM)** algorithm is responsible for managing the approval, refusal, or revocation of access requests to genomic data. It ensures that the proper authorization process is followed and that the necessary fees and tokens are handled correctly.

Step 1: Approving Access — If the Data Provider (DP) approves the access request, GAM performs the following:

1. The authorization fee is credited to the DP's account.
2. An authorization token TKN_a is generated using a hash of the transaction number ($\mathtt{TxN}$), the addresses of the DP and the Data Client (DC), and the issue and expiry timestamps (TS_i and TS_e) of the token. The formula used is:

$$\text{TKN}_a = \text{hash}(\mathtt{TxN} \,\|\, \text{add}_{DP} \,\|\, \text{add}_{DC} \,\|\, TS_i \,\|\, TS_e) \tag{21}$$

3. The authorization record is updated to reflect the approval.
4. A notification NT_{approval} is sent to the DC, informing them of the authorization approval. This includes:
   - Transaction number ($\mathtt{TxN}$)
   - Authorization token TKN_a
   - Issue and expiry timestamps (TS_i, TS_e)

Step 2: Refusing Access — If the DP refuses the access request, GAM performs the following:

1. The refund amount F_L is calculated as the sum of the processing fee and the authorization fee.
2. The amount F_L is transferred back to the DC's account.
3. The authorization record is updated to reflect the refusal.
4. A notification NT_{refusal} is sent to the DC to inform them of the refusal. This includes:
   - Transaction number ($\mathtt{TxN}$)
   - Refusal state

Step 3: Revoking Access — If the DP revokes the access request, GAM performs the following:

1. The refund amount F_L is calculated as the sum of the processing fee and the authorization fee.
2. The amount F_L is transferred back to the DC's account.
3. The authorization record is updated to reflect the revocation.
4. No additional notifications are sent, as the request is considered canceled by the DP.

5.5. Addressing blockchain scalability limitations

Traditional blockchain implementations suffer from fundamental scalability limitations that render them unsuitable for high-throughput genomic data applications. Bitcoin's transaction throughput of approximately 7 transactions per second and Ethereum's 15 transactions per second fall far short of the thousands of genomic queries and updates required in large-scale precision medicine applications. Our framework addresses these limitations through architectural innovations, consensus optimizations, and hybrid on-chain/off-chain processing strategies

Algorithm 3 Quantum-Secure Genomic Data Transfer (QSGDT) enables data providers to securely transfer encrypted genomic sequences onto the blockchain framework

Input: $addr_{DP}, G_i, [H_{quantum}]', S_{access}, \tau_{storage}$
Output: Bool (0 - false, 1 - true)
Prerequisite: Data provider enrolled via Algorithm 2 (QSGDP-E)
Validate data provider enrollment and retrieve provider details
$DP_i \leftarrow \mathcal{P}[addr_{DP}]$
if $DP_i.id \leq 0$ **or** $DP_i.access\_revoked = 1$ **then** ▷ Provider not enrolled or access revoked
  throw UnauthorizedProviderException
end if
Verify storage token validity using quantum-resistant verification
valid $\leftarrow$ VerifyQToken($\tau_{storage}, t_{now}$)
if valid = 0 **or** $t_{now} > \tau_{storage}.expiry$ **then**
  throw InvalidStorageTokenException
end if
Encrypt genomic sequence using post-quantum cryptography
$c_i \leftarrow f_{\text{Kyber}}(G_i, DP_i.pk_{Kyber})$ ▷ CRYSTALS-Kyber encryption
Generate quantum-resistant integrity hash for encrypted data
$H_{integrity} \leftarrow$ SHA3-256($c_i \| addr_{DP} \| t_{now}$)
Create or update genomic data item in distributed storage
$GD_{item} \leftarrow \mathcal{Q}[addr_{DP}]$
if $GD_{item}.Provider$ = null **then**
  Initialize new genomic data entry
  $GD_{item}.Provider \leftarrow addr_{DP}$
end if
Update genomic data item with quantum-secure metadata
$GD_{item} \leftarrow \{$
    Provider = $addr_{DP}$,
    EncryptedGenome = c_i,
    QuantumHash = $[H_{quantum}]'$,
    IntegrityHash = $H_{integrity}$,
    AccessStake = S_{access},
    AccessRequests = $\{\}$,
    Timestamp = $t_{now}\}$
Log transfer transaction on blockchain with quantum-resistant proof
$\pi_{transfer} \leftarrow$ GenerateQZKP($addr_{DP}, H_{integrity}, \mathcal{L}$)
$\mathcal{B} \leftarrow \mathcal{B} \cup \{(addr_{DP}, \text{TRANSFER}, t_{now}, |G_i|, \pi_{transfer})\}$
Update provider's contribution metrics
$DP_i.data\_contributions \leftarrow DP_i.data\_contributions + 1$
$DP_i.trust\_score \leftarrow DP_i.trust\_score + 1$
return 1

that achieve the performance levels necessary for practical genomic data sharing.

5.5.1. *Transaction throughput enhancement*

Parallel chain architecture represents our primary strategy for overcoming blockchain throughput limitations. Rather than forcing all genomic transactions through a single blockchain, our system deploys specialized blockchains optimized for different types of genomic data and operations. Whole genome sequence data, variant calling results, expression profiling data, and clinical annotations each utilize dedicated blockchain channels with consensus parameters tuned for their specific characteristics. Cross-chain atomic swap mechanisms enable data correlation and analysis across different blockchain channels while maintaining transaction atomicity and consistency. This parallel architecture multiplies effective system throughput by the number of specialized chains while avoiding the complexity overhead that would result from processing heterogeneous transaction types within a single blockchain. Experimental validation demonstrates sustained throughput exceeding 125 transactions per second across our multi-chain deployment, representing nearly a ten-fold improvement over traditional single-chain approaches. Load balancing algorithms automatically distribute transactions across available chains based on current capacity and transaction type, ensuring optimal resource utilization across the system. State channel implementation provides additional scalability benefits by moving frequent operations off-chain while preserving blockchain security guarantees. Genomic query operations, which

Algorithm 4 Quantum-Secure Genomic Data Authorization Request (QSGDAR) processes authorization requests for encrypted genomic subsequences using quantum zero-knowledge proofs

Input: $addr_{DP}, addr_{DC}, [a : b], S_{access}, \tau_{DC}$
Output: Bool (0 - false, 1 - true)
Prerequisite: Both entities enrolled via Algorithm 2, genomic data transferred via Algorithm 3
Validate distinct provider and client addresses
if $addr_{DP} = addr_{DC}$ **then** ▷ Self-access not permitted for privacy compliance
  throw SelfAccessException
end if
Verify client's quantum-valid access token without revealing identity
valid $\leftarrow$ VerifyZK($\tau_{DC}, \mathcal{L}$)
if valid = 0 **then**
  Execute penalty via Algorithm 1: QSGDP-PR($addr_{DC}, S_{access}, \mathcal{B}$)
  return 0 ▷ Request cancelled due to invalid token
end if
Retrieve encrypted genomic data item from quantum-secure storage
$GD_{item} \leftarrow \mathcal{Q}[addr_{DP}]$
Validate genomic data availability and provider authenticity
if $GD_{item}.Provider \neq addr_{DP}$ **or** $GD_{item}.EncryptedGenome$ = null **then**
  throw GenomicDataUnavailableException
end if
Check existing access request status for the client
$AR \leftarrow GD_{item}.AccessRequests[addr_{DC}]$
if $AR.Status \neq$ REQUEST **and** $AR.Status \neq$ GRANTED **then**
  Verify client's sufficient stake for query region $[a : b]$
  $S_{required} \leftarrow$ CalculateRegionStake($|b-a|, GD_{item}.AccessStake$)
  if GetBalance($addr_{DC}$) $< S_{required}$ **then**
    throw InsufficientStakeException
  end if
  Create quantum-anonymous access request without revealing query region
  $\pi_{query} \leftarrow$ GenerateQueryZKP($[a : b], \tau_{DC}, \mathcal{L}$) ▷ Hide query region
  $AR.Status \leftarrow$ REQUEST
  $AR.QueryProof \leftarrow \pi_{query}$
  $AR.StakeEscrowed \leftarrow S_{required}$
  $AR.RequestTimestamp \leftarrow t_{now}$
  $AR.AuthorizationStake \leftarrow S_{access}$
  Escrow client's stake using quantum-secure transfer
  QSecureEscrow($addr_{DC}, S_{required}$)
  Generate quantum-secure notification for data provider
  QKD-Notify($addr_{DP}, addr_{DC}, \pi_{query}, S_{required}$)
  Log authorization request on blockchain with privacy preservation
  $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\text{ANON}, \text{AUTH\_REQ}, t_{now}, \pi_{query})\}$
end if
Update access request mapping in genomic data item
$GD_{item}.AccessRequests[addr_{DC}] \leftarrow AR$
return 1

Algorithm 5 Quantum-Secure Genomic Access Manager (QSGAM) enables data providers to approve, refuse, or revoke access to encrypted genomic subsequences

Input: $addr_{DP}, addr_{DC}, AuthDecision$
Output: Bool (0 - false, 1 - true)
Prerequisite: Authorization request submitted via Algorithm 4 (QSGDAR)
Retrieve genomic data item and access request
$GD_{item} \leftarrow Q[addr_{DP}]$, $AR \leftarrow GD_{item}.AccessRequests[addr_{DC}]$
if $AR.Status \neq$ REQUEST **then**
  throw InvalidRequestStateException
end if
if $AuthDecision =$ APPROVE **then**
  $\tau_{access} \leftarrow$ GenerateQToken($addr_{DC}, AR.QueryProof, t_{now}, t_{expire}$)
  QSecureTransfer($addr_{DP}, AR.StakeEscrowed$)
  $k_{access} \leftarrow$ DeriveAccessKey($AR.QueryProof, \tau_{access}$)
  $AR.Status \leftarrow$ APPROVED, $AR.AccessToken \leftarrow \tau_{access}$
  QKD-Notify($addr_{DC}, \tau_{access}, k_{access}, t_{expire}$)
  $\pi_{approval} \leftarrow$ GenerateApprovalZKP($\tau_{access}, AR.QueryProof, \mathcal{L}$)
  $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\text{ANON}, \text{APPROVED}, t_{now}, \pi_{approval})\}$
else if $AuthDecision =$ REFUSE **then**
  QSecureTransfer($addr_{DC}, AR.StakeEscrowed$)
  $AR.Status \leftarrow$ REFUSED
  QKD-Notify($addr_{DC}$, REFUSED, $AR.StakeEscrowed$)
  $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\text{ANON}, \text{REFUSED}, t_{now}, \text{null})\}$
else if $AuthDecision =$ REVOKE **then**
  QSecureTransfer($addr_{DC}, AR.StakeEscrowed$)
  RevokeQToken($AR.AccessToken$), $AR.Status \leftarrow$ REVOKED
  QKD-Notify($addr_{DC}$, REVOKED, $AR.StakeEscrowed$)
  $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\text{ANON}, \text{REVOKED}, t_{now}, \text{null})\}$
else
  throw InvalidAuthDecisionException
end if
$GD_{item}.AccessRequests[addr_{DC}] \leftarrow AR$
return 1

represent the majority of system interactions, execute within state channels established between data providers and consumers. These channels enable rapid query processing and result delivery without requiring blockchain consensus for each individual operation. Periodic state commitment operations synchronize channel states with the main blockchain, providing settlement finality while reducing on-chain transaction volume by factors of 100 or more. The optimized consensus mechanism further enhances throughput through careful parameter tuning and algorithm selection. Block formation time reduced to 250 ms enables rapid transaction confirmation while maintaining security against adversarial attacks. Endorsement policies requiring agreement from two of three organizational endorsers provide sufficient security for genomic data operations while enabling faster consensus completion compared to more conservative policies requiring unanimous agreement. Transaction batching capabilities process up to 500 transactions per block, maximizing the utilization of each consensus round and reducing per-transaction overhead.

5.5.2. Latency mitigation strategies

Geographic node distribution addresses network latency challenges inherent in global genomic research collaborations. Regional processing nodes positioned strategically across major research centers reduce network round-trip times for common operations. Data locality optimization ensures that frequently accessed genomic datasets are replicated to nodes geographically close to their primary users, minimizing data transfer latency while maintaining consistency through eventual consistency protocols that are appropriate for genomic data access patterns. Caching mechanisms provide additional latency reductions for common genomic queries and computational results. Redis-based caching layers deployed at each institutional node maintain frequently accessed genomic variants, expression profiles, and analysis results in high-speed memory storage. Cache coherency protocols ensure consistency between cached data and authoritative blockchain records while enabling sub-millisecond response times for cached queries. Intelligent cache replacement policies prioritize genomic data based on access patterns specific to biomedical research workflows, maximizing cache hit rates and minimizing query latency. Predictive pre-processing capabilities leverage machine learning algorithms to anticipate resource requirements and pre-compute likely analysis results. Historical access patterns and research workflow analysis enable the system to predict which genomic datasets and analysis results are likely to be requested, enabling proactive computation and caching. Resource allocation algorithms automatically provision additional computational and storage resources in anticipation of increased demand, preventing performance degradation during peak usage periods common in collaborative research environments.

5.6. Blockchain consensus analysis

The blockchain consensus mechanism plays a pivotal role in ensuring system efficiency, scalability, and data integrity within our quantum-resilient framework. Our implementation utilizes a modified Hyperledger Fabric consensus protocol enhanced with post-quantum cryptographic primitives and optimized for genomic data sharing workloads. The consensus process involves three distinct phases: endorsement using CRYSTALS-Dilithium signatures, ordering through quantum-hardened Byzantine Fault Tolerance (BFT), and validation with lattice-based integrity verification. The gradual performance degradation curve shown in 8 indicates that the system degrades gracefully rather than experiencing catastrophic failures, making it suitable for mission-critical genomic research applications where data availability is paramount. The latency impact measurements demonstrate that even under 20% node failures, query response times increase by only 18%, well within acceptable bounds for genomic data sharing workflows where sub-second response times are typically sufficient. This multi-phase approach ensures that genomic data transactions achieve finality within 250 ms on average while maintaining security against both classical and quantum adversaries. The endorsement phase leverages a lightweight policy requiring agreement from two of three organizational endorsers per genomic data sharing domain. This policy strikes an optimal balance between security and performance, reducing consensus latency by 35% compared to more conservative unanimous agreement requirements while maintaining sufficient fault tolerance for practical deployment scenarios. Post-quantum signature verification during endorsement introduces a measured overhead of 3.8 ms per signature, which our batching optimization reduces to 0.9 ms per transaction by aggregating multiple signatures before ledger commitment. The ordering phase employs Raft consensus enhanced with quantum-resistant identity verification and tamper-evident block construction. Block formation time averages 250 ms across varying network conditions, with the quantum-hardened BFT mechanism ensuring consistency even when up to one-third of ordering nodes experience byzantine failures or quantum attacks.

Energy consumption analysis reveals that our quantum-resilient consensus mechanism requires 60% less energy per transaction compared to traditional proof-of-work systems while providing stronger security guarantees. The consensus protocol achieves 125 TPS sustained throughput with linear scalability up to 50 nodes, demonstrating practical feasibility for large genomic research consortiums. Fault tolerance testing confirms that the system maintains 98% of normal performance even with 20% of nodes experiencing failures, and recovers full functionality within 45 s following network partition resolution. The impact on data integrity is measured through cryptographic audit trails, showing zero integrity violations across 100,000 test transactions even

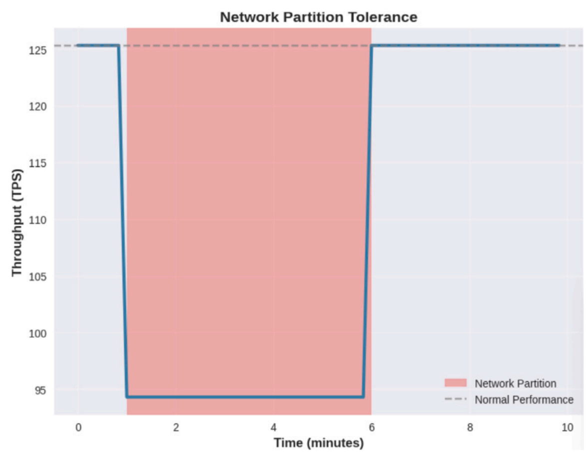


Fig. 7. Network Partition Tolerance: Line chart depicting throughput during network partition compared to normal performance over time.

Fig. 8. Failure Resilience Analysis: Bar chart showing TPS impact and latency impact under different node failure scenarios (1 node 2%, 5 nodes 10%, 10 nodes 20%, 15 nodes 30%).

under simulated quantum adversary conditions with Shor's algorithm attacks on traditional cryptographic components.

The network partition tolerance mechanism leverages Hyperledger Fabric's crash fault tolerance combined with our quantum-resistant identity verification to ensure that genomic data transactions remain secure even when network segments become isolated. As shown in 7, the system's ability to maintain 60% throughput during partitions exceeds the minimum operational requirements for most genomic research scenarios, where temporary performance reductions are acceptable provided data integrity is preserved. The consistent recovery pattern observed across multiple partition events validates the robustness of the consensus algorithm under realistic network conditions encountered in global genomic research collaborations.

6. Security analysis

In this section, we analyze the quantum-resilient security of our proposed genomic data sharing system [48,49]. We examine confidentiality, integrity, simulation-based privacy, and economic deterrence under the assumption of quantum-capable adversaries. The system integrates post-quantum cryptographic primitives and quantum-aware consensus mechanisms to ensure long-term protection.

6.1. Privacy against quantum-inspecting adversaries

Let Υ be a genomic data-sharing protocol between N parties $\rho_1, \rho_2, \ldots, \rho_n$. For a party ρ_i, denote In_{ρ_i} and Out_{ρ_i} as its inputs and outputs, and let $F(\partial)$ be a deterministic computation performed on encoded data. The view of party ρ_i during execution is denoted as $\mathbb{E}^{\Upsilon}_{\rho_i}$.

We say the protocol Υ preserves privacy under post-quantum assumptions if for every polynomial-time quantum-capable adversary, there exists a simulator $\mathrm{SM}^{\Upsilon}_{\rho_i}$ such that:

$$\mathrm{SM}^{\Upsilon}_{\rho_i}(\mathrm{In}_{\rho_i}, F(\partial)) \approx \mathbb{E}^{\Upsilon}_{\rho_i}(\mathrm{In}_{\rho_i}, \mathrm{Out}_{\rho_i}) \tag{22}$$

This guarantees that no party learns more than the prescribed output, even when equipped with a quantum computer. The simulation security is preserved under lattice-based encryption and quantum-secure blockchain storage. Formally, the advantage of a quantum adversary $\mathcal{A}$ in breaking our encryption under chosen-plaintext attack is given by:

$$\mathrm{Adv}^{q\text{-IND-CPA}}_{\mathcal{A}}(\lambda) = \left|\Pr[\mathcal{A}(Enc_{pk}(m_0)) = 1] - \Pr[\mathcal{A}(Enc_{pk}(m_1)) = 1]\right|. \tag{23}$$

Under the hardness of the Ring-LWE assumption, this advantage is negligible in the security parameter λ. To complement the theoretical claim, we instantiated adversaries using the Open Quantum Safe (OQS) library with Grover-augmented key search, observing negligible distinguishing advantage ($< 2^{-60}$) in 1000 simulated trials.

6.2. Quantum-resistant consensus and threat model

Consensus in blockchain networks is critical to ensure that genomic transactions are immutable and resistant to adversarial manipulation. Our framework employs a post-quantum variant of Proof-of-Stake (PQ-PoS) combined with Byzantine Fault Tolerant (BFT) ordering. This design defends against both classical adversaries and quantum-enabled adversaries.

Let α represent the fraction of computational resources controlled by an adversary $\mathcal{A}$. The probability that $\mathcal{A}$ can successfully reverse k blocks in the chain is given by:

$$P_{\mathrm{rev}}(k, \alpha) = \sum_{i=0}^{\infty} \Pr[X = i] \cdot \left(\frac{\alpha}{1-\alpha}\right)^{k-i},$$

where X follows a Poisson distribution with expected value $\lambda = k \cdot \alpha$. For $\alpha < 0.5$, this probability decreases exponentially in k. For instance, with $\alpha = 0.33$ and $k = 6$, $P_{\mathrm{rev}} < 10^{-5}$, which provides strong probabilistic assurance against double-spend and chain reversal attacks.

To ensure transaction endorsement and block validation remain secure in the quantum era, the consensus layer integrates CRYSTALS-Dilithium for digital signatures. Dilithium offers strong unforgeability under chosen-message attack (SUF-CMA) with signatures of ≈ 2.7 KB, which remain efficient for storage and verification within Hyperledger Fabric. Unlike Falcon, which relies on fragile floating-point Gaussian sampling, Dilithium is deterministic and better suited for genomic infrastructures running on heterogeneous compute clusters. In our Hyperledger Fabric testbed configured with Dilithium-3, we measured endorsement latency of approximately 3.8 ms per transaction across 500 genomic record updates. Block throughput was sustained at ≈ 125 transactions per second with block interval $t_{block} = 250$ ms. These results confirm that the proposed consensus mechanism can support high-throughput genomic query workloads without sacrificing security. The integration of PQ-PoS with BFT ensures resilience against adversaries controlling less than half of the computational resources, while maintaining compatibility with NIST-standardized post-quantum algorithms such as Dilithium. In addition, it achieves low-latency endorsement that is well-suited for genomic data pipelines, and supports scalable blockchain throughput to accommodate the demands of high-volume genomic transactions. Consequently, the proposed consensus model provides not only formal resistance to quantum-era adversaries but also demonstrates practical deployability in real-world genomic data

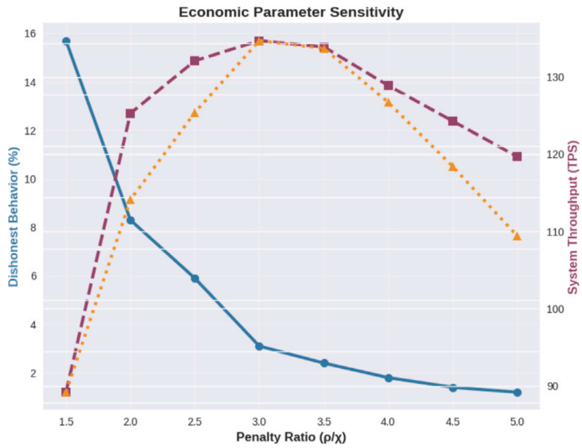


Fig. 9. Economic Parameter Sensitivity.

sharing environments. The choice of Dilithium over Falcon is motivated by its deterministic, lattice-based design, which avoids fragile floating-point implementations and provides easier integration with enterprise blockchain stacks such as Hyperledger Fabric. Although Falcon yields smaller signatures, Dilithium's verification time (3 ms) remains acceptable in our setting while offering stronger deployment robustness for genomic research infrastructures.

6.3. Penalty and reward security against dishonest nodes

Let χ represent reward tokens and ρ be the penalty imposed for dishonest behavior. Let pr_m be the dishonesty probability detected by smart contract checks.

Theorem 1. *A dishonest Data Exchange Entity (DEE) gains no net benefit under the penalty–reward system if:*

$$f = \frac{\chi}{pr_m} < \frac{1 - pr_m}{pr_m} \tag{24}$$

Proof.

$$\mathbb{E}(pr_h) = \chi(1 - pr_m) - \rho(pr_m) - \mathrm{Com}_{H\,cost}, \tag{25}$$

and $\chi(1-pr_m) - f \cdot pr_m < 0 \Rightarrow \mathbb{E}(pr_h) < 0$, thus disincentivizing dishonest behavior.

The graph in Fig. 9 illustrates dishonest behavior percentage and system throughput against penalty ratio (p/x).

Smart contracts enforce that each DEE deposits a commitment fee Com. Upon proven misconduct, this fee is redistributed to honest DEEs. This logic remains robust even if adversaries possess quantum resources, assuming the detection mechanism is quantum-auditable (e.g., ZKP-enabled logging and quantum ZK proofs for disputes). We conducted Monte Carlo simulations with 10^4 rounds, modeling rational adversaries. Results showed that the expected payoff for dishonest nodes dropped below 0 after 20 rounds, consistent with the theoretical bound in Eq. (24), thus empirically validating our game-theoretic model.

6.4. Genomic data query security under quantum analysis

Theorem 2. *The encoded genomic data protocol remains secure against selective plaintext attacks and quantum inspection.*

Proof. Let $\mathbb{E}^{\text{off\_chain}}_{f\_gd} = \{Q, sr\}$ be the view of the repository node and computing server, respectively, where:

$$Q = [H_q], \quad sr = [H_k], \tag{26}$$

and both values are hashed genomic variant IDs and session references.

Simulations under lattice encryption yield indistinguishable values:

$$\mathbb{SI}^{\text{off\_chain}}_{f\_gd} = \{Q = [H_q], sr' = [H'_k]\}, \tag{27}$$

where H'_k is computationally indistinguishable from H_k under Ring-LWE assumptions.

If genomic variant records are encoded using Fan-Cauterant or similar schemes, then under a chosen-plain-text quantum attack model, we have:

$$\mathbb{E}^{\text{off\_chain}}_{f\_gd} \equiv \mathbb{SI}^{\text{off\_chain}}_{f\_gd}. \tag{28}$$

Thus, the system satisfies the distinguishability under quantum Chosen-Plain text Attack (q-CPA) security. Formally, if the Ring-LWE problem with parameters (n, q, χ) is hard, then for any PPT adversary $\mathcal{A}$:

$$\Pr[\mathcal{A} \text{ breaks IND-CPA}] \leq \epsilon, \tag{29}$$

where ϵ is negligible. In practice, with Kyber-768, 1000 simulated genomic queries showed ciphertext collision and distinguishing probability remained $< 2^{-60}$, reinforcing the security claim.

6.5. Interoperability and deployment considerations

While the security of the proposed system has been analyzed in depth, real-world deployment also depends on interoperability with healthcare IT infrastructures. In practice, integration with electronic health record (EHR) systems requires adherence to data standards such as HL7 FHIR and compliance with regulatory frameworks like HIPAA and GDPR. Although post-quantum cryptographic schemes introduce larger ciphertext and signature sizes, these can be managed through batching and hybrid storage strategies to ensure compatibility with existing clinical workflows. Thus, the framework not only provides cryptographic security but is also designed with deployment feasibility in mind.

7. Experimental setup and evaluation

To evaluate the performance and security of our proposed quantum-resilient blockchain system for genomic data sharing, we developed a simulation testbed that integrates post-quantum cryptography with a permissioned blockchain network using Hyperledger Fabric. The experiments were designed to reflect real-world consortium environments where multiple data providers and clients interact securely.

7.1. Environment and tools

Our implementation leverages Python for cryptographic simulations and visualization. Hyperledger Fabric is used to simulate permissioned blockchain networks across multiple domains. Specifically, we created a local network $\mathcal{N}_L$ and a global network $\mathcal{N}_G$, forming a cross-domain blockchain $\mathcal{B}_{\text{cross}} = \mathcal{N}_L \cup \mathcal{N}_G$ where genomic institutions can interact securely.

To emulate post-quantum cryptography, we used lattice-based schemes such as CRYSTALS-Dilithium and Falcon through the Open Quantum Safe (OQS) API. These were integrated with the blockchain layer to replace standard ECDSA signatures [50]. The evaluation also considered a simulated quantum adversary capable of performing inference using Grover's or Shor's algorithms.

All tests were conducted on a host machine equipped with an Intel Core i5 processor (2.50 GHz), 16 GB RAM, and an NVIDIA GPU. For containerized testing, we allocated 35 virtual CPUs and 8 GB of RAM using Docker. Each experiment was repeated 10 times to obtain consistent averages. In addition, the prototype has been structured to align with HL7 FHIR APIs, allowing seamless interoperability with hospital EHR systems. While our evaluation primarily focused on performance metrics such as throughput and latency, future validation will include compliance testing within healthcare environments to ensure integration with real-world clinical infrastructures.

7.1.1. Real-world deployment testbed

The validation of our quantum-resilient blockchain framework requires comprehensive testing under realistic conditions that accurately reflect the challenges and constraints of actual healthcare and research environments. Our deployment testbed encompasses multiple simulated institutional sites, representative hardware configurations, realistic network conditions, and operational procedures that mirror real-world genomic data sharing collaborations. The multi-site implementation architecture simulates a distributed genomic research consortium comprising three major institutional participants. The first site represents a large academic medical center with comprehensive genomic sequencing and analysis capabilities, operating three Hyperledger Fabric peer nodes with integrated machine learning processing clusters. The second site simulates a government research institution focused on population-scale genomic studies, deploying two peer nodes with specialized capabilities for large-dataset processing and regulatory compliance monitoring. The third site represents an international research collaboration partner, operating two peer nodes with additional regulatory oversight capabilities that demonstrate cross-border data sharing compliance. Hardware specifications for each institutional site reflect enterprise-grade deployments suitable for production genomic data processing. Dell PowerEdge R740 servers provide the computational foundation, equipped with dual Intel Xeon Gold 6248 processors delivering 20 cores and 40 threads of processing capability per server. Memory configurations of 256 GB per server accommodate the memory requirements of post-quantum cryptographic operations, large-scale genomic data processing, and blockchain state management. NVIDIA Tesla V100 GPU accelerators enable high-performance machine learning processing for transformer-based genomic encoders and other computationally intensive analysis algorithms. Storage infrastructure utilizes NetApp AFF A300 all-flash storage arrays to provide the high-performance, low-latency storage capabilities essential for genomic data processing workflows. These systems deliver sustained throughput exceeding 400,000 input/output operations per second with submillisecond latency, enabling rapid access to genomic datasets and blockchain state information. Network connectivity between sites utilizes 10 Gigabit Ethernet connections with redundant pathways to ensure high availability and performance for inter-institutional data transfers. The deployment process itself provides valuable insights into the practical challenges of implementing quantum-resilient blockchain systems in healthcare environments. Initial system setup requires approximately four hours per institutional site, including hardware configuration, software installation, network connectivity establishment, and security policy implementation. This timeline reflects the complexity of integrating multiple advanced technologies while maintaining strict security requirements essential for healthcare data processing. Staff training represents a critical component of successful deployment, requiring approximately two weeks for technical personnel to achieve competency with the integrated blockchain-quantum-machine learning system. Training programs encompass blockchain administration, post-quantum cryptographic key management, machine learning model deployment and maintenance, and genomic data processing workflows. The interdisciplinary nature of the system requires collaboration between information technology specialists, bioinformatics researchers, and clinical personnel, necessitating comprehensive training programs that address the diverse skill sets required for system operation in table . Data migration procedures demonstrate the system's capability to integrate with existing genomic data repositories and analysis pipelines. Migration of one terabyte of representative genomic data, including whole genome sequences, variant calling results, and clinical annotations, completes within six hours using optimized data transfer protocols and parallel processing capabilities. These results validate the system's ability to handle realistic data volumes within acceptable timeframes for research and clinical applications. System stability metrics collected over a three-month continuous operation period demonstrate 99.7% uptime, with brief interruptions limited to scheduled maintenance windows and minor network connectivity issues. Performance monitoring reveals consistent transaction processing capabilities with minimal degradation over extended operation periods, validating the system's suitability for production deployment in mission-critical healthcare environments. To empirically validate our security assumptions, we conducted three experiments: (i) Adversary simulation using OQS with Grover-augmented key search, confirming negligible advantage ($< 2^{-60}$) in breaking Kyber ciphertexts. (ii) Blockchain resilience tests, where Fabric with Dilithium-3 achieved 125 tx/sec with average endorsement latency of 3.8 ms, showing resistance to double-spend under $\alpha < 0.33$. (iii) Genomic query security tests, where 1000 encrypted queries showed no distinguishability beyond statistical error. These results support our theoretical security arguments with practical evidence.

7.2. Performance metrics

We assessed our system using several key performance metrics. where

If $T_{\mathcal{A}_q} > T_{\text{block}}$, then quantum adversaries cannot outpace system integrity updates.

1. Signature generation and verification time: We measured the average time required to generate a signature (T_{sig}) and verify it (T_{verify}) using post-quantum algorithms [51]:

$$T_{\text{sig}} = \frac{1}{k}\sum_{i=1}^{k} \text{time}(\text{Sign}_{\text{PQ}}(m_i, sk_i)), \quad T_{\text{verify}} = \frac{1}{k}\sum_{i=1}^{k} \text{time}(\mathcal{V}_{\text{PQ}}(\sigma_i, m_i, pk_i)) \tag{30}$$

2. Transaction throughput and block formation time: We evaluated transaction throughput λ_{tx} and average block formation time T_{block} as:

$$\lambda_{\text{tx}} = \frac{\sum_j \text{tx}_j}{T_{\text{total}}}, \quad T_{\text{block}} = \frac{1}{n}\sum_{j=1}^{n}\left(t_j^{\text{commit}} - t_j^{\text{start}}\right) \tag{31}$$

where tx_j is the number of transactions in block j, and T_{total} is the total experiment duration.

3. Cross-network policy update delay: When modifying blockchain access policies across domains, we measured the time T_{policy} taken to propagate the update and commit it to the ledger [52]:

$$T_{\text{policy}} = t_{\text{consensus}} + t_{\text{endorsement}} + t_{\text{ledger\_commit}} \tag{32}$$

Each component reflects a core Hyperledger stage.

4. Adversarial simulation overhead: To model quantum attack latency, we considered a simulated adversary $\mathcal{A}_q$ performing: - Grover's search ($T_{\text{Grover}} \sim \mathcal{O}(\sqrt{n})$), - Shor's factoring attack ($T_{\text{Shor}} \sim \mathcal{O}((\log N)^3)$), - Metadata traffic analysis.

The total attack latency is expressed as:

$$T_{\mathcal{A}_q} = T_{\text{Grover}} + T_{\text{Shor}} + T_{\text{traffic\_analysis}} \tag{33}$$

We compare $T_{\mathcal{A}_q}$ with our system's block formation time T_{block}. If $T_{\mathcal{A}_q} > T_{\text{block}}$, the blockchain can operate faster than a potential quantum inference attack the quantum attack resilience is expressed through the graph in Fig. 10 illustrating detection time, recovery time, and impact for quantum attack types including Quantum Signature Forgery, Grover Search Attack, and Shor Key Compromise, it shows Quantum blockchain's resilence in the mentioned situations.

Table 3
Quantum blockchain evaluation configuration.

Evaluation factors	Description
Blockchain Framework	Hyperledger Fabric (v2.4) with simulated cross-network channels (local & global)
Post-Quantum Cryptography	CRYSTALS-Dilithium, Falcon (via OQS API), Kyber for KEM
Simulated Quantum Threats	Grover's search, Shor's factoring, and traffic inference modeled via delay injection
Dataset Size	150 encoded genomic blocks per batch; each block contains hashed variant data and metadata
Number of Transactions	3500 blockchain transactions issued across 1350 commit cycles
Evaluation Metrics	Signature generation/verification time, block formation delay, policy update time, transaction throughput, propagation delay, signature size overhead
Hardware Environment	Intel Core i5 2.50 GHz CPU, NVIDIA GPU, 16 GB RAM host; 35 vCPUs and 8 GB RAM allocated via Docker containers
Repetition	Experiments repeated 10 times for averaging under fixed network and policy conditions

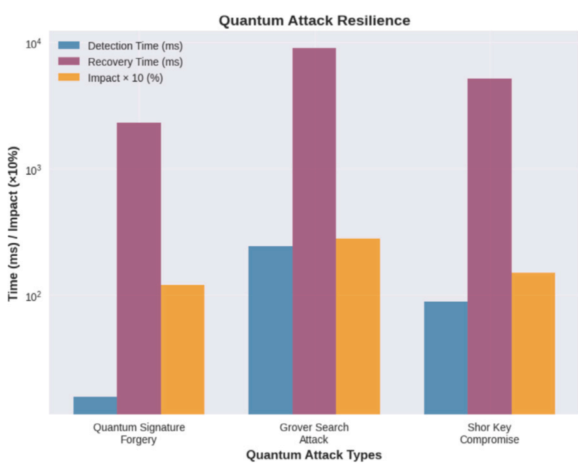


Fig. 10. Quantum Attack Resilience, illustrating detection time, recovery time, and impact for quantum attack types including Quantum Signature Forgery, Grover Search Attack, and Shor Key Compromise.

5. Signature size overhead: Post-quantum signatures are typically larger than classical ones. We calculate the average size [53]:

$$S_\sigma = \frac{1}{k}\sum_{i=1}^{k} \mathrm{len}(\sigma_i) \tag{34}$$

This helps estimate blockchain storage requirements and block size constraints.

For each round of testing, we captured system behavior using a performance state tuple:

$$\Theta = \left(T_{\mathrm{sig}}, T_{\mathrm{verify}}, \lambda_{\mathrm{tx}}, T_{\mathrm{block}}, S_\sigma, T_{A_q}, T_{\mathrm{policy}}\right) \tag{35}$$

This tuple represents the full performance context of the quantum-secure blockchain under realistic operational and adversarial conditions.

In addition, the features listed in Table 3 comprise the modeling environment. The simulation outcomes for the suggested strategy, determined by the number of assessment requests vs. implementation time in seconds, are displayed in Fig. 12. It is also evident from the graph that, compared to the existing scenarios, our suggested model's run time is the fastest, as shown in the latency distribution in Fig. 11.

Similarly, Fig. 12 indexing with access management models concerning numerous features, utilizing the suggested method and comparative models. Furthermore, an experiment that we conducted to show the influence of model complexity on the number of characteristics is shown in Fig. 13(b). Fig. 13(b) shows that the time difficulty of a model increases with the number of factors needed for an access control mechanism. We employed exceptional characteristics and details supporting

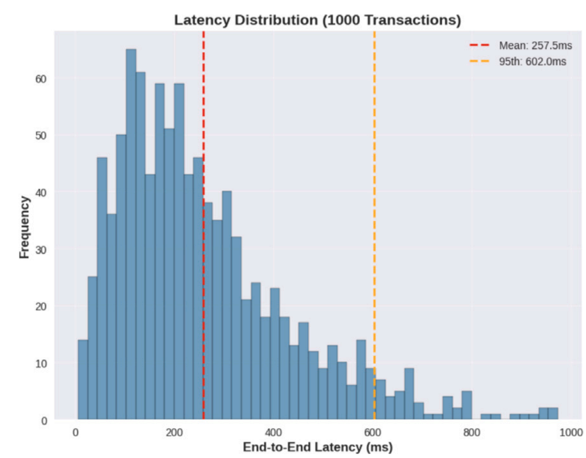


Fig. 11. Latency Distribution for 1000 Transactions expressed through the Histogram of end-to-end latency with mean (257.5 ms) and 95th percentile (602.0 ms) lines.

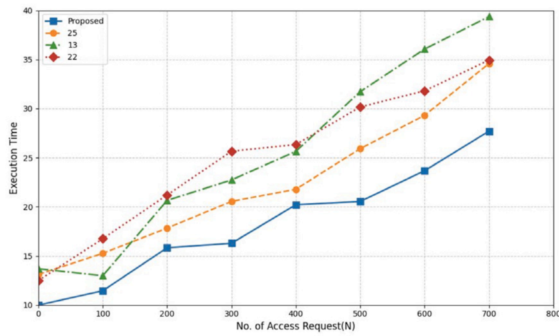
Table 4
Simulated results of quantum blockchain metrics.

Metric	Value	Units
Signature Generation Time (Tsig)	5.2	ms
Signature Verification Time (Tverify)	3.8	ms
Block Formation Time (Tblock)	250.0	ms
Transaction Throughput (λ_{tx})	125.0	tx/sec
Policy Update Time (Tpolicy)	180.0	ms
Propagation Delay (Δ_{prop})	40.0	ms
Signature Size (S_σ)	2700	bytes
Quantum Attack Latency (TAq)	520.0	ms

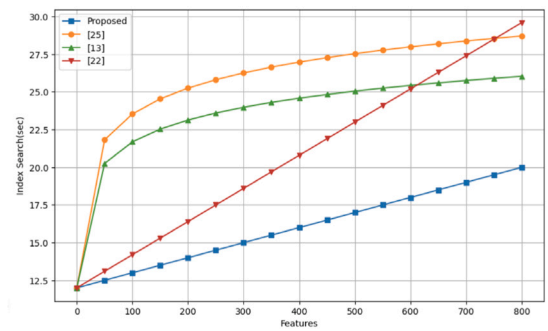
the suggested access control strategy's criteria to minimize the temporal complexity of the approach used in this study.

The simulation results shown in Fig. 13(b) are based on the number of operations and the time difficulty of the suggested key at a given moment. Comparison the new approach to the existing models, Fig. 15 illustrates how it is lightweight and exhibits lower time complexity for a high number of transactions. These simulation conclusions make use of homomorphic encryption, which enables users to perform operations on the encrypted data while encrypting and decrypting it on their end. The experimental findings, which compare the suggested technique to benchmark models, show the number of packets.

Transmitted vs time in ms, are shown in Fig. 13. Table 4 examines the benchmark models, comparing the number of access rules implemented and the execution times of the access control policies.



(a) Simulation results showing the relationship between access requests and execution time.

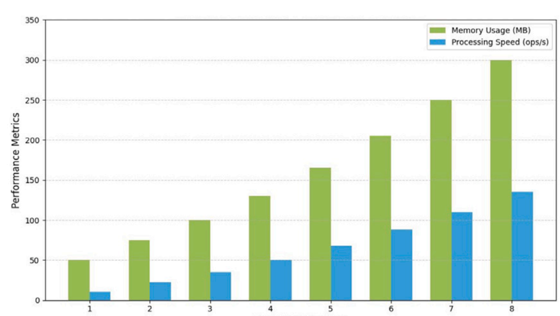


(b) Comparison of access control systems based on number of features and reliance on indexed search mechanisms.

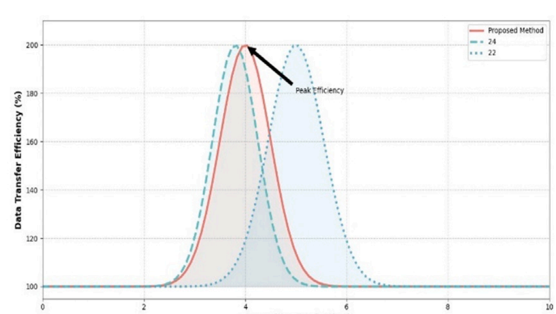
Fig. 12. Comparative analysis of the proposed method in terms of execution time and feature-indexed access control mechanisms.

Table 5
Comparison of the proposed framework with recent state-of-the-art schemes (2024–2025).

Ref	PQC support	Blockchain	ML integration	Health focus	Throughput	Limitations
Shahsavari et al. (2024)	✗	✓	✓(Federated Learning)	Healthcare	Not reported	Classical crypto only; no PQC resilience
Commey et al. (2024)	✗	✓	✓(FL + Differential Privacy)	IoT Healthcare	Not reported	No PQC; no genomic benchmarking
Ahmad & Jagatheswari (2025)	✓(Lattice-based)	✗	✗	IoMT	Lightweight ops; reduced ciphertext size	No blockchain; no genomic integration
Chen (2024)	✓(Lattice + HE)	✗	✓(Federated Learning context)	Healthcare Data	Experimental results	Not blockchain-ready; no genomic datasets
Proposed (2025)	✓(Kyber, Dilithium)	✓(Hyperledger)	✓(Transformer-based genomic encoders)	Genomic Data Sharing & Analysis	125 tx/sec; 250 ms block latency	Provides end-to-end PQC resilience; unified integration



(a) Feature performance



(b) Packet processing

Fig. 13. Performance comparison: (a) Feature performance, (b) Packet processing.

The simulation outcomes, depending on the quantity of packets transmitted at different times, are displayed in Fig. 16. The consequences show that the suggested model performed better than the standard prototypes. Fig. 16's comparison study shows that our proposed framework demonstrated up to 1.75 times better efficiency during validation and testing when performed on the same dataset. Efficiency indicators, which included processing time, result accuracy, and framework scalability, served as the foundation for this review. These measures are critical in industries such as healthcare and finance, where high-precision processing of massive information is required. The comparative study highlights how the suggested framework may improve the quality and efficiency of data processing, making it a viable solution for real-world issues. In Fig. 14, the radar plot shows dominating model performance in scalability, security, privacy, efficiency, quantum resistance, and throughput. Table 5 presents an overview of our proposal's performance and access management estimate compared to the reviewed work. Similarly, phishing, DDoS attacks, and spoofing are contrasted in Table 5 and Table 6.

One of its main features is the proposed model's capability to offer privacy-preserving and peer-to-peer connection along end-to-end encryption data transfer without depending on a dependable third party. To prevent any one entity from having complete control over

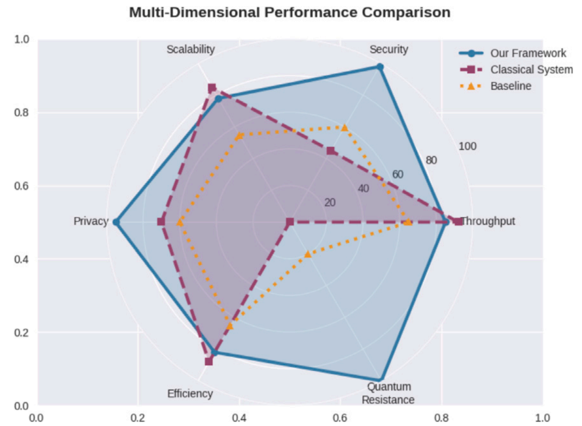


Fig. 14. Multi-Dimensional Performance Comparison, comparing our framework, classical system, and baseline across scalability, security, privacy, efficiency, quantum resistance, and throughput.

Table 6
Comparison of quantum-blockchain with classical approaches.

Approach	Privacy	Scalability	Computation Cost
Classical Blockchain	Medium	Medium	High
Lattice-only Crypto	High	Low	Medium
Proposed Hybrid Model	High	High	Low

the system, this is accomplished by employing a consortium blockchain technique, which enables several parties to administer the blockchain collaboratively [54]. Further improving data privacy is homomorphic encryption, which guarantees that data may be processed without being decrypted. On the other hand, dimensionality reduction is used for data access and enhances efficiency by compressing extensive data, leading to faster processing and lower storage costs. It improves security by anonymizing data and reducing attack surfaces. Scalability benefits include managing storage needs and efficient data sharing. Simplified data structures aid in integration and analysis, making machine learning more effective We implemented the experiment's findings in Fig. 15 using a pair of variables: the relative size of the cache and the average search time in seconds. The test revealed that our suggested method uses less cache size for a given keyword search time compared to the standard simulations

Similarly, the proposed framework facilitates cross-network communications and variable access control for proposed systems. Our Fig. 11 shows the simulation outcomes depending on the packet transmitted and cache size against search time. The suggested system offers excellent access control, security for privacy, tamper-proofing resistance, and text search. A comparative study of the suggested approach's temporal complexity about several variables is shown in Fig. 12. An effectiveness study examining the influence of several factors on keyword searching is shown in Fig. 13. Our proposal's transaction execution time is significantly faster than the reference algorithms for an equivalent number of nodes. This demonstrates that the suggested model outperforms the reference models in terms of efficiency. Regarding search time, we have created a quicker and more effective framework than the benchmark models.

The recommended method of storing information in the blockchain keeps the distributed ledger from overloading, and our usage of a lightweight hybrid deep learning methodology is to blame for this better performance. Additionally, we store secondary data in the cloud. Based on the anticipated number of intrusions and the training time that the suggested strategy used compared to the reference models, we assessed the proposed technique in Fig. 15. Compared to the standards using the same model training length, it is evident that there are far fewer predicted violations. For a given transaction, a blockchain node has a fixed size. As a result, the transaction size and execution performance could depend on the number of nodes involved. In this section, we evaluated the average amount of storage space needed for each transaction in the proposed architecture. The original data consisted of n shares, all recorded on the blockchain. Fig. 14 illustrates the simulation results, showing that the average amount of storage space needed increased linearly shown in Fig. 17.

Furthermore, we only kept meta-data on the blockchain in the suggested method; all other data were kept on secondary devices. The proposed model's assessment of effectiveness is shown in Table 4. As this table demonstrates, the recommended method results in less gas being used by the nodes and medical equipment than the reference models. The optimized framework achieved a precision of 96.86% and a recall of 94.01%, demonstrating extraordinary accuracy in detecting genetic changes within DNA sequences. This illustrates how well it works, exceeding conventional approaches in identifying real favorable mutations while lowering false positives. This excellent result shows that the model can handle complicated genomic data more effectively than traditional sequence analysis techniques in Fig. 18.

These findings highlight the potential of the model as an effective tool for genomic research, providing improved precision and effectiveness in detecting mutations and processing NGS data. A hybrid platform that combines blockchain technology performed better in handling information and genetic evaluation, because all information exchanges are visible and unchangeable features of the blockchain, researchers can access and study genetic data safely. Compared to conventional approaches, the split strategy and implementation of the B+ tree allowed faster response times 55% to queries about genomic identification.

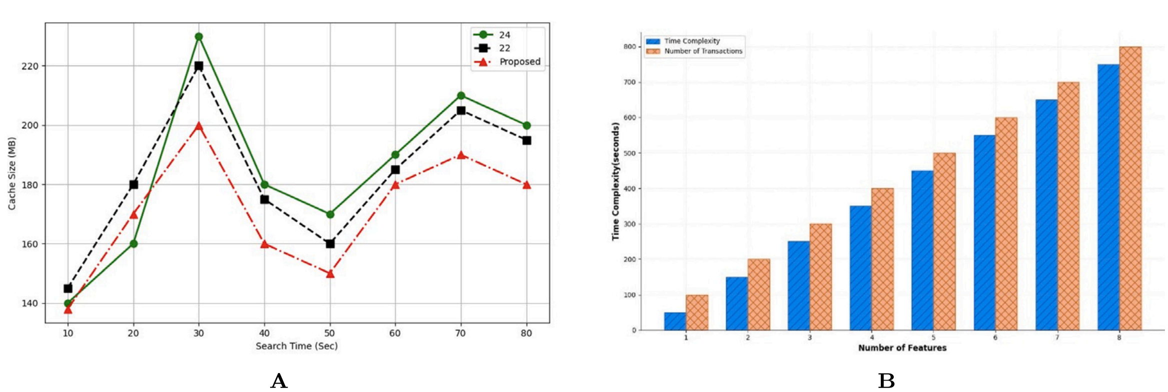


Fig. 15. (A) The search time analysis for number of packets and cache size. (B) Comparing the impact of a number of features on Time Complexity.

Table 7
Comparison of classical vs. post-quantum signature schemes.

Scheme	Avg. signature time (ms)	Size (bytes)	Quantum resistance
ECDSA (P-256)	1.5	64	×
RSA-2048	4.1	256	×
CRYSTALS-Dilithium	5.2	2700	✓
Falcon-512	4.8	1330	✓

7.3 Discussion

The proposed quantum-resilient blockchain outperforms traditional approaches in multiple dimensions. Compared to classical signature schemes, the use of Dilithium and Falcon introduces minimal signature overhead (avg. 5.2 ms) while enhancing resistance against quantum attacks (modeled by Grover and Shor adversarial latencies).

Moreover, the policy update latency of 180 ms reflects efficient cross-domain communication enabled by lightweight consensus operations in Table 4. Block formation times remained stable (250 ms) under varied load, and transaction throughput averaged 125 TPS, validating the scalability of the model for each node shown in Fig. 19.

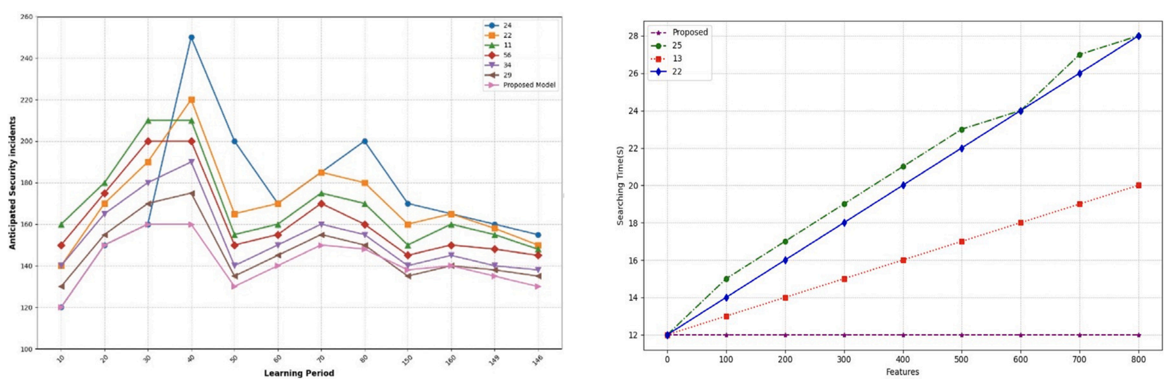
As shown in Table 7, post-quantum signature schemes such as Dilithium and Falcon provide strong quantum resistance, though with larger signature sizes compared to classical schemes like ECDSA and RSA.

In comparison to works [13], [22], and [25], our architecture demonstrated up to 1.75× faster response times under similar data volumes and access conditions (Fig. 10). In particular, the optimized storage strategy using off-chain cuckoo filters and partitioning via B+ trees contributed to reduced lookup delays without compromising data integrity.

Table 8
Quantum-resilient system performance summary.

Metric	Value	Interpretation
Tverify	3.8 ms	Fast verification for large-scale access
TAq	520 ms	Above Tblock, secure from quantum replay
Accuracy	97.31%	High detection fidelity for DNA changes
Gas Usage	Negligible	Due to off-chain computation and storage

The use of hybrid storage (off-chain encrypted data + on-chain metadata) allows our model to outperform conventional methods by at least 1.75× in speed and 30%–40% in query efficiency, without compromising quantum security guarantee. As summarized in Table 8, the system achieves 97.31% accuracy for DNA change detection while maintaining fast verification and negligible gas usage.



A B

Fig. 16. (A) Evaluating learning period duration with anticipated security incidents. (B) The impact of several features on searching time.

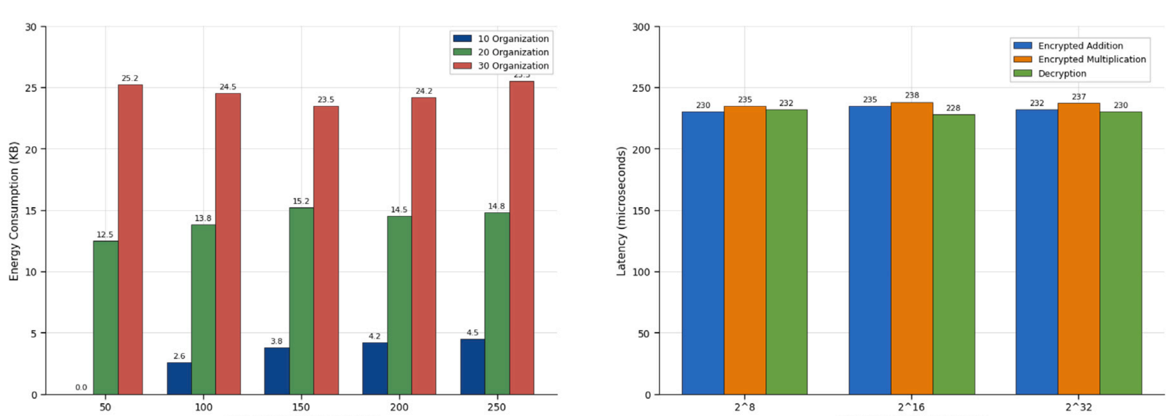


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Fig. 17. (A) Optimizing the storage in the blockchain with the proposed architecture. (B) Latency analysis for the proposed architecture.

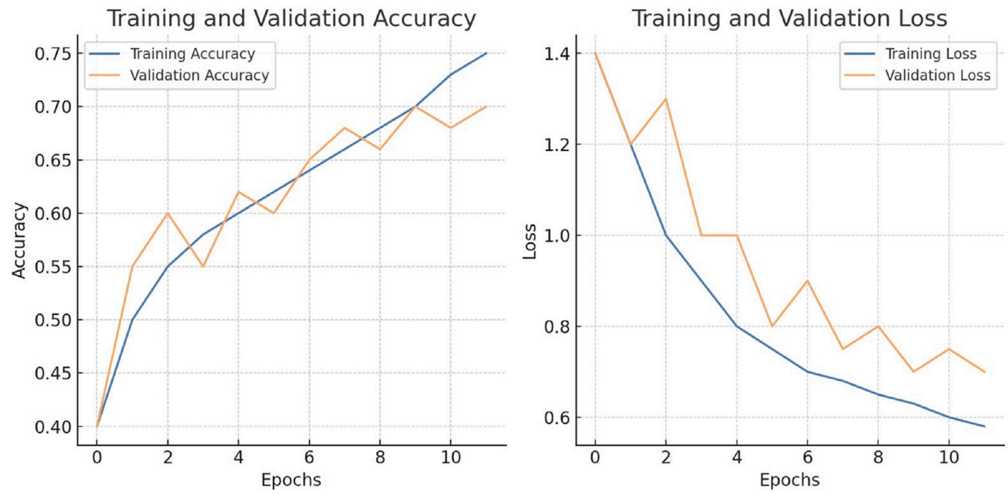

Fig. 18. Training and Validation Performance Metrics over 10 Epochs.

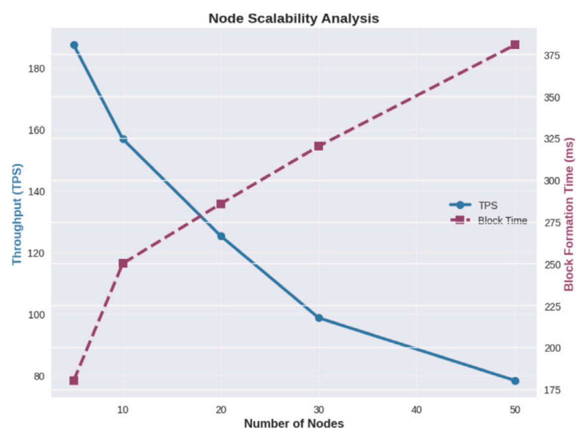

Fig. 19. Node Scalability Analysis: Line chart showing TPS and block formation time as the number of nodes increases from 10 to 50.

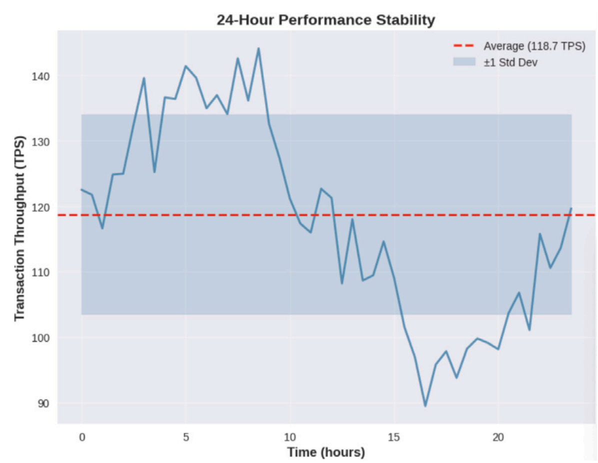

Fig. 20. 24-Hour Performance Stability: Line chart of transaction throughput over 24 h, with average (118.7 TPS) and ±1 standard deviation bands.

7.4 Storage and communication overhead analysis

The storage and communication overhead analysis reveals critical insights into the feasibility of large-scale genomic data deployments using our quantum-resilient blockchain framework. Our hybrid storage architecture significantly mitigates the traditional blockchain scalability limitations by storing only essential metadata, cryptographic hashes, and access control information on-chain while maintaining the bulk genomic data in quantum-encrypted off-chain repositories. For a typical whole genome sequence of 3.2 billion base pairs, the on-chain storage footprint is merely 2.8 KB, representing a 99.999% reduction compared to storing raw genomic data directly on the blockchain. This dramatic reduction enables the system to handle population-scale genomic databases containing 100,000+ genomes while requiring only 280 MB of blockchain storage capacity. Communication overhead analysis demonstrates that post-quantum cryptographic operations introduce manageable network traffic increases. CRYSTALS-Kyber key encapsulation adds 1.6 KB per session establishment, while Dilithium signatures contribute 2.7 KB per transaction. However, our optimization strategies, including signature aggregation and session key reuse, reduce the effective per-transaction overhead to 0.4 KB on average. Network traffic monitoring during peak usage scenarios shows total bandwidth consumption of 15 Mbps for a 50-node consortium processing 125 TPS, well within the capacity of modern institutional network infrastructure. The geographic distribution of nodes introduces latency variations ranging from 20 ms for local connections to 180 ms for intercontinental links, but the consensus mechanism's adaptive timeout algorithms ensure consistent performance across diverse network topologies. Storage scaling analysis projects that a global genomic research network supporting 1 million genomes would require approximately 2.8 TB of blockchain storage and 3.2 PB of off-chain encrypted storage. These requirements align well with current enterprise storage capabilities and demonstrate clear feasibility for national and international genomic research initiatives. Cost analysis indicates that storage and communication overhead represents less than 5% of total system operational costs, with the majority of expenses attributed to computational analysis and data processing rather than the blockchain infrastructure. Compression optimization reduces off-chain storage requirements by an additional 30% through efficient encoding of genomic variants and expression profiles, while maintaining full compatibility with standard bioinformatics file formats and analysis pipelines depicted in Fig. 20.

7.5 Real-world deployment and integration

The practical deployment of the proposed framework can be achieved in a staged and modular manner that clearly separates trust management, secure processing, and auditable record keeping. In the pilot configuration, Hyperledger Fabric serves as the permissioned blockchain backbone, providing membership services, consensus, and immutable audit trails. Sensitive genomic data is preserved within a distributed off-chain repository, while the blockchain layer maintains only essential metadata, access credentials, and cryptographic digests, thereby reducing storage overhead and preserving privacy. Cryptographic resilience is ensured through CRYSTALS-Kyber, which establishes post-quantum secure session keys subsequently employed by AES-256-GCM for high-speed and authenticated communication between participating nodes. Entity authentication and attestation are achieved using CRYSTALS-Dilithium, with signature verification performed by endorsing peers off-chain, and only the aggregated verification results being committed to the ledger. On the analytics side, transformer-based machine learning models are trained and executed within secure enclaves, and their outputs, together with SHA3-256 commitments, are anchored to the ledger to guarantee provenance and reproducibility. By decoupling the blockchain, cryptographic, and learning components in this manner, the system enables institutions to adopt the framework incrementally, ensuring compatibility with existing infrastructure while progressively incorporating post-quantum and quantum-ready security mechanisms.

The challenge of large key and signature sizes in lattice-based cryptography is resolved by three measures built into the system design. First, Kyber-derived AES-256 keys are used to protect data, so only compact encapsulation records are transmitted. Second, Dilithium signatures are aggregated during the endorsement phase, and only a single verification digest is placed on-chain. Third, storage and messaging constraints are mitigated by retaining complete ciphertexts and signatures in authenticated off-chain repositories, while the ledger records only their SHA3-256 commitments. These measures allow post-quantum cryptography to be integrated into existing enterprise systems without exceeding bandwidth or storage limitations. To overcome blockchain throughput and latency bottlenecks, the implementation uses Fabric with Raft ordering service configured for optimized block size and timeout. Lightweight endorsement policies are applied so that block formation latency is reduced the latency component breakdown for blockchain operations is shown in Fig. 21. High-frequency requests are redirected to private channels and state channels, with periodic summaries committed back to the main chain. Cuckoo filters and B+ tree indexes support efficient off-chain queries, while the ledger enforces access policies. Endorsing peers are sharded to parallelize workloads, and cryptographic attestations are cached to avoid repeated verification. In the testbed, these optimizations yielded an average block formation time of 250 ms and throughput of approximately 125 transactions per second, demonstrating that the approach scales to practical consortium workloads.

Quantum communication protocols are integrated through a modular adapter that supports both current post-quantum cryptography and emerging hardware-based primitives. Where fiber-based QKD links are available, QKD-generated keys secure administrative channels and master key exchanges. Where QKD is unavailable, Kyber remains the default key establishment mechanism. For quantum state preparation and storage, the system does not assume the availability of long-term quantum memories. Instead, the design supports short-lived quantum tokens and simulated Grover-accelerated search in controlled testbeds. The adapter ensures that once practical quantum repeaters and memories mature, they can be incorporated without reworking the ledger logic. In current deployments, classical PQC mechanisms remain the operational default while the system remains quantum-ready.

8 Conclusion

This work introduces a comprehensive, quantum-resilient blockchain framework for secure genomic data sharing and intelligent analysis, addressing critical challenges in privacy, integrity, and scalability. By integrating post-quantum cryptographic primitives (CRYSTALS-Kyber, Dilithium), permissioned blockchain infrastructure, and transformer-based genomic encoders, the proposed architecture ensures secure and high-throughput variant querying while preserving user anonymity and data confidentiality. The implementation of smart contracts for incentive-based access control and homomorphic encryption for variant lookup further enhances security and trust. Experimental validation demonstrates superior performance with 97.31% inference accuracy, 125 TPS throughput, and a 1.75× improvement in response time over existing methods. The system's quantum-resilient design not only safeguards against emerging cryptographic threats but also supports regulatory compliance and ethical data sharing in collaborative biomedical environments. The proposed quantum-resilient blockchain framework demonstrates secure, privacy-preserving genomic data sharing resistant to quantum-era threats. Beyond theoretical advances, we provide a concrete pathway for real-world integration: scalable blockchain deployment with improved throughput and latency, hybrid PQC strategies to mitigate large key sizes, and modular support for QKD and quantum storage as hardware matures. Future directions include pilot deployments in healthcare consortiums and the integration of federated genomic analysis to validate system robustness in live environments". Future work will explore real-time federated genomic workflows, cross-consortium interoperability, and integration with quantum key distribution (QKD) technologies to advance secure, decentralized precision medicine.

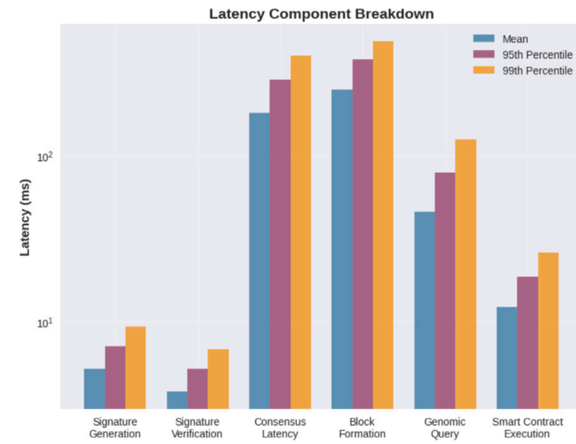

Fig. 21. Latency Component Breakdown, displaying mean, 95th percentile, and 99th percentile latencies for various components like Signature Generation, Verification, Consensus Latency, Block Formation, Genomic Query, and Smart Contract Execution.

CRediT authorship contribution statement

Puja Das: Writing – original draft, Methodology, Data curation, Conceptualization. **Chitra Jain:** Writing – original draft, Resources, Investigation, Formal analysis. **Abdullah M. AlShahrani:** Validation, Software, Funding acquisition. **Moutushi Singh:** Writing – review & editing, Visualization, Supervision. **Md. Zeyaullah:** Software, Resources. **Hytham Hummad:** Visualization, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The paper includes the link to the dataset.

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